Succinct Shortness Check Under a Few Kilobytes via Tensor Train Random Projections

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Abstract. Range proofs are a fundamental building block of lattice-based proof systems. Existing approaches relying on standard Johnson-Lindenstrauss (JL) struggle to provide succinctness: unstructured JL incurs linear verifier complexity, while structured JL introduced in RoK and Roll [ASIACRYPT'25] produce large projection vectors that need to be sent in costly committed form. For a witness of dimension m and a security parameter λ, the JL projection vector is $\mathcal{O}(\lambda)$, but the projection matrix is $\mathcal{O}(\lambda m)$. Structured JL can be represented in $\widetilde{\mathcal{O}}(\rho\lambda^2)$, but the resulting projection length grows to $\widetilde{\mathcal{O}}(m/\rho)$ for any trade-off parameter $\rho > 1$.

In this paper, we introduce Tensor Train Random Projection (TTRP) to lattice-based cryptography, offering a "best of both world" structured JL variant. The Tensor Train (TT) matrix representation drops to $\widetilde{\mathcal{O}}(\lambda)$, while maintaining the projection vector at $\mathcal{O}(\lambda)$.

We show that TTRP can be seamlessly integrated into proof systems by linearizing the well-formedness of the projection with the sum-check protocol. Concretely, using TTRP + sum-check as a drop-in replacement for existing range proof techniques yields an approximate shortness check with $\mathcal{O}(\lambda \log m (\operatorname{loglog} m)^2)$ verifier complexity and proof size around 4 KB for relevant parameter settings.

Keywords: Lattices · Tensor Train · Approximate Range Proofs

1 Introduction

At the core of lattice-based succinct arguments is an opening proof of an Ajtai commitment [Ajt96], i.e a proof of knowledge of a witness $\mathbf{w} \in \mathcal{R}^m$ such that

$$\mathbf{A}\mathbf{w} = \mathbf{t} \bmod q \tag{1}$$

$$\|\mathbf{w}\| < B, \tag{2}$$

where $\mathcal{R} := \mathbb{Z}[X]/(f(X))$ is a cyclotomic polynomial ring and $\mathbf{A}$ is a uniformly random matrix over $\mathcal{R}_q := \mathcal{R}/q\mathcal{R}$. In particular, it is not enough to prove that $\mathbf{w}$ satisfies the linear constraint eq. (1), one also needs to convince the verifier that the witness is short.

However, constructing shortness proofs and extracting the corresponding witness are notoriously difficult. Typically, the extractor of a lattice-based proof system recovers a so-called *relaxed opening* of an Ajtai commitment, consisting of a tuple $(\bar{\mathbf{w}}, s)$ such that

$$\mathbf{A}\bar{\mathbf{w}} = \mathbf{t} \bmod q \tag{3}$$

$$\|s\bar{\mathbf{w}}\| < B. \tag{4}$$

Rather than proving both eqs. (1) and (2), the extractor obtains a $\bar{\mathbf{w}}$ with an unconstrained norm satisfying $\mathbf{t} = \mathbf{A}\bar{\mathbf{w}} \bmod q$, and a factor s with a small operator norm such that $s\bar{\mathbf{w}}$ is small. The extracted witness is typically constructed as a quotient

$$\bar{\mathbf{w}} := \mathbf{v}s^{-1} \bmod q$$

where $\mathbf{v} \in \mathcal{R}^m$ is a vector with a slightly larger norm bound $\|\mathbf{v}\| \leqslant B'$ than B and $s := c - c'$ is a challenge difference.

This extraction slack s is a consequence of a fundamental incompatibility between the following desirable requirements on the challenge set $\mathcal{C}$:

(i) For completeness, any challenge $c \in \mathcal{C}$ must have a small operator norm so that our commitment scheme is linearly homomorphic w.r.t coefficients in $\mathcal{C}$. In other words, $c\mathbf{w}$ must remain small enough that $c\mathbf{t}$ is a binding commitment.
(ii) For soundness, any extraction slack factor $s := c - c'$ for $c, c' \in \mathcal{C}$ must have an inverse, and its inverse must have a small operator norm. Should there exist a small bound ϱ such that $\|s^{-1}\|_{\mathsf{op}} \leqslant \varrho$, our extractor could extract $\bar{\mathbf{w}}$ with a mildly looser norm bound $\|\bar{\mathbf{w}}\| \leqslant \varrho B'$.
(iii) For soundness, $|\mathcal{C}| = 2^{\mathcal{O}(\lambda)}$ where λ is the security parameter. Since our soundness arguments inherently rely on the Schwartz-Zippel Lemma, a sub-exponential-size challenge set would require costly repetitions.

From a high level, [AL21] proved that, in cyclotomic rings, there exists no challenge set that satisfies (i), (ii) and (iii). The relaxed-opening strategy consists in choosing a challenge set $\mathcal{C}$ of polynomials with bounded coefficients that satisfy 1) and 3), but not 2). More precisely, we can only ensure that the inverse exists modulo q [LS18], but its operator norm can be arbitrarily large. The *subtractive sets* strategy explored in [ACK21,CLM23,KLNO24] consists in choosing a challenge set that satisfies (i) and (ii) of maximal size $\mathsf{poly}(\lambda)$.

While relaxed openings can be sufficient for constructing simple primitives such as signatures [Lyu09,Lyu12,DDLL13,DKL+18], the existence of an extraction slack factor can be a source of inefficiency for more advanced primitives. For example, existing lattice-based Succinct ARguments of Knowledge (SNARKs), Polynomial Commitment Schemes (PCS) or folding schemes rely on recursive Reductions of Knowledge (RoK), where the operator norm of the extraction slack grows exponentially with the recursive depth. The prevailing strategy to bridge the gap between a relaxed opening and an exact solution to eqs. (1) and (2) is to attach a *range proof*[3].

Infinity norm proofs. Earlier constructions of linear-size proofs focused on infinity-norm proofs [YAZ+19,BLS19,ESLL19,LNS21], i.e, proving claims of the form

$$\|\mathbf{w}\|_\infty \leqslant B_\infty \tag{5}$$

for some infinity norm bound B_∞. With respect to the coefficients of $\mathbf{w}$ over $\mathbb{Z}$, the eq. (5) can be expressed as a polynomial and therefore proven with protocols relying on Schwartz-Zippel arguments. More recently, folding scheme constructions [NS26,BC25b,GLLO26,BC25a] adapted this infinity-norm proof strategy to the succinct setting. In those schemes, the polynomial relations encoding eq. (5) are batched together and reduced to a single random polynomial evaluation via the sum-check protocol. With the exception of [BC25b], the polynomial encoding eq. (5) for all aforementioned linear and succinct proofs is of degree $2B_\infty + 1$, which corresponds to the range of the coefficients of the witness.

With today's tools, proving high-degree polynomial relations over lattices is expensive; therefore, the most efficient regime for these infinity-norm proofs is when $B_\infty = 1$[4]. The consequence is that witnesses need to be decomposed to ternary coefficients and possibly re-committed in this form, which is a major source of inefficiency. Moreover, the infinity norm is less tightly controlled than the Euclidean norm during naturally occurring operations such as linear combinations, and therefore infinity norm proofs tend to yield worse concrete parameters than Euclidean norm proofs.

Random projections and Euclidean norm proofs. First introduced in functional analysis, Johnson-Lindenstrauss [JL84] (JL) projections map a high-dimensional vector $\mathbf{w}$ to a smaller space

[3] For succinct proofs with a recursive structure, the rigorous name should be range RoK.
[4] Again, with the exception of LatticeFold+'s range proof [BC25b], though at the price of expensive double commitments.

via a linear transformation of the form $\mathbf{p} = \mathbf{J}\mathbf{w}$. JL projections are widely adopted and studied in machine learning [Ach03,BM01] as a dimensionality-reduction tool.

In the context of lattice-based proofs, random projections were explored in the setting of amortization [BL17], and fully connected to Johnson-Lindenstrauss projections in [GHL22]. Random projections are philosophically radically different from Schwartz-Zippel-based infinity-norm proofs, as they keep the geometric nature of the norm bound (instead of arithmetizing it as a polynomial) and rely on a statistical test to distinguish long vectors from short vectors.

At a high level, the idea is that with a witness vector $\mathbf{w} \in \mathbb{Z}^m$ and a projection matrix $\mathbf{J} \in \mathbb{R}^{k \times m}$ whose coefficients are drawn from $\mathcal{N}$ (here $\mathcal{N}$ is the centered Gaussian distribution of standard deviation 1), the distribution of $\|\mathbf{J}\mathbf{w}\|^2$ follows a scaled chi-squared distribution with k degrees of freedom concentrated around $k\|\mathbf{w}\|^2$. For any vector $\mathbf{w} \in \mathbb{Z}^m$, there exist a lower bound γ_1 and an upper bound γ_2 that only depend on a security parameter λ (or confidence parameter) and the height k of the projection matrix such that

$$\Pr_{\mathbf{J} \leftarrow \mathcal{N}^{k \times m}} \left(\gamma_1 \leqslant \frac{\|\mathbf{J}\mathbf{w}\|}{\sqrt{k}\|\mathbf{w}\|} \leqslant \gamma_2 \right) \geqslant 1 - 2^{-\lambda}.$$

Discretizing $\mathbf{J}$ to $\mathbb{Z}^{k \times m}$ by sampling the coefficients from the distribution $\mathcal{D}$ such that $\mathcal{D}(1) = \mathcal{D}(-1) = 1/4$ and $\mathcal{D}(0) = 1/2$, we obtain a compact projection vector $\mathbf{p} := \mathbf{J}\mathbf{w} \in \mathbb{Z}^k$ of small dimension $k = \mathcal{O}(\lambda)$ that, intuitively, is small if and only if the original witness $\mathbf{w}$ is small.

Wrapping up, to include a JL approximate range proof for a vector $\mathbf{w}$ such that $\|\mathbf{w}\| \leqslant B$ in a lattice-based proof system, the verifier generates a random matrix $\mathbf{J}$ with entries drawn from $\mathcal{D}$, and the prover sends the projection $\mathbf{p}$ together with a proof of well-formedness that $\mathbf{p} = \mathbf{J}\mathbf{w}$. The verifier checks that $\|\mathbf{p}\| \leqslant \gamma_2 B$, which holds with overwhelming probability for an honest prover. Conversely, the extractor knows that for all $\mathbf{w} \in \mathbb{Z}^m$, if $\|\mathbf{w}\| \geqslant \frac{\gamma_2}{\gamma_1} B$, then

$$\Pr_{\mathbf{J} \leftarrow \mathcal{D}^{k \times m}} (\|\mathbf{J}\mathbf{w}\| \leqslant \gamma_2 B) \leqslant 2^{-\lambda},$$

thus achieving an approximate ℓ_2-norm proof with a *slack factor* $\frac{\gamma_2 B/\gamma_1}{B} = \frac{\gamma_2}{\gamma_1}$. For practical proof systems [LNP22,BS23], the concrete slack factor is $\sqrt{128/30} \simeq 2$, yielding a tight ℓ_2-norm proof.

Random projections for succinct arguments. The challenge of a JL approximate range proof is a $k \times m = \mathcal{O}(\lambda m)$ ternary matrix, which means that schemes such as LaBRADOR [BS23] that use straightforward JL projections suffer from a linear verifier[5]. To patch this issue, [KLNO25,KLN+26] introduced so-called *structured random projections*. Their idea is to divide the witness into chunks of a given length ℓ, and to use a single JL matrix to project each chunk individually. While this technique does lower the verifier complexity to $\mathcal{O}(\lambda\ell)$, several problems arise.

First, the projection is now a vector $\mathbf{p}$ of m/ℓ projections, each of length $\mathcal{O}(\lambda)$, which is too large to be sent in the clear. Their solution is to instead commit to $\mathbf{p}$, which means that the prover must now convince the verifier that $\mathbf{p}$, sent in committed form, is small. The second problem is that the optimal parameter ℓ chosen by `RoKoko` is $\mathcal{O}(\lambda^2)$, making the overall verifier complexity at least $\mathcal{O}(\lambda^3)$ per recursion.

1.1 Our contributions

In this paper, we introduce Tensor Train Random Projections (TTRP) to the design of lattice-based proof systems. TTRP is the first random projection to simultaneously achieve the following three properties:

[5] We note that despite relying on JL projections, Greyhound [NS24] achieves $\mathcal{O}(\lambda\sqrt{m})$ verifier complexity due to calling LaBRADOR on inputs of length $\mathcal{O}(\sqrt{m})$

(i) Succinct representation in $\widetilde{O}(\lambda)$[6].
(ii) Projection length linear in $\mathcal{O}(\lambda)$.
(iii) Slack factor in $\mathsf{poly}(\lambda, m)$.

Our contributions can be divided in two parts.

Theoretical groundwork. While the properties of TTRP have been extensively studied by the machine learning community, the statistical bounds in the settings relevant to the design of approximate range proofs remained unexplored. We bridge this gap by showing that TTRP is a viable approximate-isometry tool for lattice-based cryptography.

First, we extend the study of TTRP to prime fields $\mathbb{Z}_q$ to fit the native arithmetic of lattice-based cryptography. We show that–much like standard JL projections–when working modulo some prime q, the approximate-isometry guarantees of TTRP only hold for a bounded range of input norms. In other words, there is a maximum norm for the witness $\mathbf{w}$ after which the statistical control on the projection collapses.

Second, we derive new ℓ_2-concentration bounds that yield practical slack factors for lattice-based proof systems. Asymptotically, these concentration bounds ensure that TTRP satisfies

(i) Succinct representation in $\mathcal{O}(\lambda \log m (\mathrm{loglog}\, m)^2) = \widetilde{O}(\lambda)$.
(ii) Projection length in $\mathcal{O}(\lambda)$.
(iii) Slack factor $\mathsf{poly}(m)$.

Notably, we provide assumption-free bounds for TTRP with entries following the distribution $\mathcal{D}$. Schemes that rely on JL projections with entries following $\mathcal{D}$ adopt the heuristic assumption that $\mathcal{D}$ is a Gaussian distribution, in which case the JL bounds are standard chi-square tail bounds. In contrast, our analysis is generic to any distribution that is a linear combination of Rademacher variables (i.e uniform over $-1, 1$) and thus does not rely on any heuristic.

Approximate range RoK. We describe a norm check protocol Π^{TTRP} that is a self-reduction for a general Ajtai commitment polynomial relations. This protocol is an approximate norm-check module in the sense that the extractor only needs relaxed witnesses to recover the input witness, though with a norm bound multiplied by the slack factor of TTRP.

This norm check RoK consists in following the JL blueprint, but relies on the sum-check protocol to linearise the well-formedness of the projection, drawing inspiration from recent works [KLOT25,GLLO26,NS26,BC25b,KLN+26]. As a self-RoK for general Ajtai commitment relations, Π^{TTRP} can be used as a highly efficient drop-in replacement for JL, unstructured JL, or sum-check ternary proofs and serve as range proof module.

To substantiate the broad applicability of our techniques, we provide examples of integration to state-of-the-art lattice-based succinct arguments. More precisely, we use our protocol as a replacement of JL in LaBRADOR, thus achieving a $\sim 90,000\times$ improvement in verifier complexity at the cost of a ~ 2 KB increase in proof size and a $\sim 2^9\times$ increase in the slack factor (Table 1). We substitute both the coarse and fine unstructured JL in `RoKoko` with our protocol, which allows us to remove costly commitments to projection vectors, and reduce the verifier complexity of the random projection by a factor $\mathcal{O}(\lambda^2/\log^2 m)$. In particular, the bottleneck for verifier efficiency in the resulting construction is now the shrink factor. Finally, we replace the sum-check based ternary proofs in Cyclo with TTRP, thus removing expensive decompositions and extension commitments. The resulting folding proof size drops by 30%.

[6] For asymptotics, we assume that $m = \mathsf{poly}(\lambda)$

Range proof	$m\varphi$	Proof Size (KB)	Slack Factor	Matrix Repr. (MB)
JL (From LaBRADOR)	2^{27}	1.84	$\sqrt{128/30}$	6492
	2^{30}	1.89	$\sqrt{128/30}$	51936
TTRP (this paper)	2^{27}	3.90	$\sim 2^{10}$	0.2
	2^{30}	3.98	$\sim 2^{10}$	0.6

Table 1. Comparison of range proof efficiency between standard JL (from LaBRADOR) and our Π^{TTRP} protocol (TTRP + sum-check linearization, fig. 1).

1.2 Technical overview

The foundational problem that we address in this paper is to find a class of projection matrices with approximate-isometric properties and succinct representation. To understand the technical aspects of our construction, let us first explore the naive approaches.

Naive approaches. A natural starting point is to consider a projection matrix where each row is a tensor product of small JL vectors. Specifically, we assume that $m = d^\mu$, and create for all $i \in [k]$ vectors $\mathbf{v}_i = \bigotimes_{j=1}^{\mu} \mathbf{a}_j^{(i)}$ for independently distributed random vectors $\mathbf{a}_j^{(i)} \leftarrow \mathcal{D}^d$. For each row of the resulting matrix $\mathbf{v}_i$, we can apply recursively the single-row JL bound and obtain a slack factor of $\mathcal{O}(\alpha^\mu)$. Due to high correlation between the output coefficient of each row, the per-row probability drops dramatically, requiring a prohibitive $\Omega(\lambda \log m)$ repetitions to achieve λ bits of statistical soundness.

Another approach is to recursively apply the structured JL projection from [KLNO25]. More precisely, the verifier would generate μ JL matrices $\mathbf{J}_1, \ldots, \mathbf{J}_\mu$ and compute the projection $\mathbf{p}$ of $\boldsymbol{w}$ as follows:

$$\mathbf{p} := \mathbf{J}_1(\mathbf{J}_2 \otimes \mathbf{I}_{kd}) \cdots (\mathbf{J}_\mu \otimes \mathbf{I}_{kd^{\mu-1}})\mathbf{w}^{7}.$$

The length ℓ of the JL projections must be at least kd, so that at each layer, the length of the projected vector is reduced by a constant factor d. At each layer, the slack factor is multiplied by α, thus this approach achieves a slack factor $\alpha^\mu = \mathcal{O}(m)$. However, the representation of the projection matrix is $\mu k^2 d = \mathcal{O}(\lambda^2 \log m)$, which is too large to be satisfactory for us. Both aforementioned approaches are particular extreme cases of our TTRP approach.

Introduction to TTRP. The Tensor Train (TT) structure was first introduced in [Ose11] as a numerical tool to represent high-dimensional data in a compact format. More recently, TTRP have been developed as an efficient dimensionality-reduction tool for machine learning [RR21,RR20].

A TT is a Kronecker product of so-called *core tensors*. Specifically, consider a 3-rd order tensor $\mathcal{M} \in \mathbb{Z}^{c_0 \times d \times c_1}$. Our 3-rd order tensor $\mathcal{M}$ can be seen as a list of $c_0 \times c_1$ matrix *slices*, and for all $j \in [d]$, we denote $\mathcal{M}(j)$ the j-th matrix slice of $\mathcal{M}$. Consider a second 3-rd order tensor $\mathcal{M}' \in \mathbb{Z}^{c_1 \times d' \times c_2}$, where the first dimension c_1 of $\mathcal{M}'$ is identical to the 3-rd dimension of $\mathcal{M}$. We define the Kronecker product $\mathcal{M} \otimes \mathcal{M}'$ as

$$\forall (i,j) \in [d] \times [d'], \quad \mathcal{M} \otimes \mathcal{M}'(i,j) = \mathcal{M}(i) \cdot \mathcal{M}'(j) \in \mathbb{Z}^{c_0 \times c_2}. \tag{6}$$

With words, the Kronecker product of $\mathcal{M}$ and $\mathcal{M}'$ is the 3-rd order tensor of dimension $\mathbb{Z}^{c_0 \times (d \times d') \times c_2}$ that is the tensor product of the vector of slices of $\mathcal{M}$ with the vector of slices of $\mathcal{M}'$. Generally,

[7] This projection strategy was recently explored in concurrent work [Osa26]

the TT of a list of μ cores $\mathcal{M}_1 \in \mathbb{Z}^{c_0 \times d \times c_1}, \ldots, \mathcal{M}_\mu \in \mathbb{Z}^{c_{\mu-1} \times d \times c_\mu}$ is

$$\mathsf{TT}(\{\mathcal{M}_i\}_{i\in[\mu]}) := \mathcal{M}_1 \otimes \cdots \otimes \mathcal{M}_\mu \in \mathbb{Z}^{c_0 \times d^\mu \times c_\mu}.$$

At a high level, the idea of TTRP is to construct a projection matrix whose rows have a tensor product structure for succinctness, but that carry more information than straightforward tensor products of constant-size JL rows. Specifically, to ensure that a TT outputs a single row vector, we enforce $c_0 = c_\mu = 1$ and define the internal rank of the TT as $c := c_1 = \cdots = c_{\mu-1}$. This internal rank intuitively plays the role of an internal repetition rate, allowing the final vector to carry more information.

TTRP is a random projection with a matrix whose rows are TT of independent lists of cores, and is given by

$$f^k_{\mathsf{TT}(\mu,\mathsf{c})}(\mathbf{w}) := \left(\mathsf{TT}(\{\mathcal{M}_i^{(j)}\}_{i\in[\mu]})_{j\in[k]}\right)\mathbf{w}.$$

Over the integers, TTRP satisfies the ℓ_2-approximate-isometry property, that is:

$$\frac{\|f_{\mathsf{TT}}(\mathbf{w})\|_2^2}{\|\mathbf{w}\|_2^2} \in (1 \pm \epsilon)$$

with high probability for some $\epsilon > 0$. Our modular ℓ_∞ concentration bound states that for any $\boldsymbol{w}$ such that $\|\boldsymbol{w}\|_\infty \geqslant B_\infty$ for some $B_\infty < q/4$, we have

$$\Pr\left[\|f^k_{\mathsf{TT}(\mu,c)}\|_\infty < \frac{B_\infty}{2^\mu}\right] \quad \leqslant \quad \left(1/2 + \frac{\mu-1}{2^{c+1}}\right)^k.$$

To achieve $\mathcal{O}(\lambda)$ repetitions, we need $\frac{\mu-1}{2^{c+1}} = \mathcal{O}(1)$, i.e $c = \mathcal{O}(\log\mu) = \mathcal{O}(\mathrm{loglog}\, m)$ thus achieving $\mathcal{O}(\lambda \log m(\mathrm{loglog}\, m)^2)$ representation for the projection matrix. Looking towards applications, we extend these modular concentration bounds to the ℓ_2-norm, where the asymptotic bounds require slightly worse parameters than in the ℓ_∞-norm case. Specifically, since the projection coefficients do not follow a Gaussian distribution, its Euclidean norm is not as concentrated as the chi-square distribution of regular JL. Concretely, we obtain approximate-isometry concentration bounds in ℓ_2-norm for $k = \mathcal{O}(\lambda)$ and $c = \mathcal{O}(\log m)$. The representation of this TTRP projection matrix is asymptotically $\mathcal{O}(\lambda \log^3 m) \in \widetilde{\mathcal{O}}(\lambda)$.

Proving TTRP. The random projection paradigm for approximate range proof is as follows. The verifier sends a projection matrix, the prover computes the projection, and sends it to the verifier together with a well-formedness argument.

Drawing inspiration from [KLN+26], we design a self-RoK for a general polynomnial relation for Ajtai commitments. Specifically, we follow the JL approximate range proof framework, and use the sum-check protocol to create a simple, succinct and cheap linearization proof for the well-formedness of the TTRP projection $\mathbf{p} \in \mathbb{Z}^k$ of a vector $\mathbf{w} \in \mathcal{R}_q^{m/\varphi}$ over a power-of-two cyclotomic ring of degree φ.

2 Preliminaries

Notation. $[n] := \{1, 2, \ldots, n\}$. We denote column vectors by bold lowercase letters, matrices by bold uppercase letters and higher-order tensors by calligraphic uppercase letters (e.g., $\mathcal{M} \in \mathbb{Z}^{c_1 \times d \times c_2}$ for 3rd-order tensor). For $i \in [d]$, we use $\mathcal{M}(i) \in \mathbb{Z}^{c_1 \times c_2}$ to denote the i-th slice of $\mathcal{M}$. We define an operator $\mathsf{vec}(\cdot)$ which flattens a matrix into a vector by *vertically* concatenating its columns. We define $\mathsf{Mat}(\cdot)$ which flattens a 3rd-order tensor into a matrix by *horizontally* concatenating its matrix slices.

Kronecker Product (slices level). Let $\mathcal{A} \in \mathbb{Z}^{c_1 \times d_1 \times c_2}$ and $\mathcal{B} \in \mathbb{Z}^{c_2 \times d_2 \times c_3}$. We define their Kronecker product, denoted by $\mathcal{A} \otimes \mathcal{B}$, as a new 3rd-order tensor $\mathcal{C} \in \mathbb{Z}^{c_1 \times (d_1 \cdot d_2) \times c_3}$. The matrix slices of $\mathcal{C}$ are generated by the standard matrix multiplication of the respective component slices

$$\mathcal{C}(i,j) = (\mathcal{A} \otimes \mathcal{B})(i,j) := \mathcal{A}(i) \cdot \mathcal{B}(j) \in \mathbb{Z}^{c_1 \times c_3},$$

where $(i,j) \in [d_1] \times [d_2]$ is the multi-index.

Example. Let $\mathcal{A} = \begin{bmatrix}\begin{bmatrix}1 & 2\end{bmatrix} \begin{bmatrix}3 & 4\end{bmatrix}\end{bmatrix} \in \mathbb{Z}^{1\times 2\times 2}$ and $\mathcal{B} = \begin{bmatrix}\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix} \begin{bmatrix}5 & 6\\ 7 & 8\end{bmatrix}\end{bmatrix} \in \mathbb{Z}^{2\times 2\times 2}$, then we have

$$\begin{aligned}\mathcal{A} \otimes \mathcal{B} &= \begin{bmatrix}\begin{bmatrix}1 & 2\end{bmatrix} \cdot \begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix} \begin{bmatrix}1 & 2\end{bmatrix} \cdot \begin{bmatrix}5 & 6\\ 7 & 8\end{bmatrix} \begin{bmatrix}3 & 4\end{bmatrix} \cdot \begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix} \begin{bmatrix}3 & 4\end{bmatrix} \cdot \begin{bmatrix}5 & 6\\ 7 & 8\end{bmatrix}\end{bmatrix}\\ &= \begin{bmatrix}\begin{bmatrix}7 & 10\end{bmatrix} \begin{bmatrix}19 & 22\end{bmatrix} \begin{bmatrix}15 & 22\end{bmatrix} \begin{bmatrix}43 & 50\end{bmatrix}\end{bmatrix} \in \mathbb{Z}^{1\times 4\times 2}.\end{aligned}$$

and consequently, we have:

$$\mathsf{Mat}\,(\mathcal{A} \otimes \mathcal{B}) = \begin{bmatrix}7 & 10 & 19 & 22 & 15 & 22 & 43 & 50\end{bmatrix} \in \mathbb{Z}^{1\times 8}.$$

The Kronecker product for 3rd-order tensors and the Kronecker product for vectors has the following mixed product similarity.

Fact 1. *For any 3rd-order tensors $\mathcal{A} \in \mathbb{Z}^{c_1 \times d_1 \times c_2}$ and $\mathcal{B} \in \mathbb{Z}^{c_2 \times d_2 \times c_3}$, it holds that:*

$$\mathsf{Mat}(\mathcal{A} \otimes \mathcal{B}) = \mathsf{Mat}(\mathcal{A}) \cdot (\mathbf{I}_{d_1} \otimes \mathsf{Mat}(\mathcal{B})) \in \mathbb{Z}^{c_1 \times (c_3 d_1 d_2)} \tag{7}$$

where $\mathbf{I}_{d_1}$ is the $d_1 \times d_1$ identity matrix.

Proof. By rewriting the right-hand side, we have

$$\begin{aligned}\mathsf{Mat}(\mathcal{A}) \cdot (\mathbf{I}_{d_1} \otimes \mathsf{Mat}(\mathcal{B})) &= \mathsf{Mat}\big(\begin{bmatrix}\mathcal{A}(1) & \dots & \mathcal{A}(d_1)\end{bmatrix}\big) \begin{bmatrix}\mathsf{Mat}(\mathcal{B}) & \mathbf{0} & \dots & \mathbf{0}\\ \mathbf{0} & \mathsf{Mat}(\mathcal{B}) & \dots & \mathbf{0}\\ \vdots & \vdots & \ddots & \vdots\\ \mathbf{0} & \mathbf{0} & \dots & \mathsf{Mat}(\mathcal{B})\end{bmatrix}\\ &= \mathsf{Mat}\big(\begin{bmatrix}\mathcal{A}(1) \cdot \mathsf{Mat}(\mathcal{B}) & \dots & \mathcal{A}(d_1) \cdot \mathsf{Mat}(\mathcal{B})\end{bmatrix}\big)\\ &= \mathsf{Mat}(\mathcal{A} \otimes \mathcal{B}).\end{aligned}$$

Rings. For a power of two φ and a positive integer q, let $\mathcal{R}$ and $\mathcal{R}_q$ denote the rings $\mathbb{Z}[X]/(X^\varphi + 1)$ and $\mathbb{Z}_q[X]/(X^\varphi + 1)$, respectively. For a polynomial $x \in \mathcal{R}$, we denote by $\mathsf{cf}(x)$ its coefficient vector in $\mathbb{Z}^\varphi$. We use $\sigma_{-1} \in \mathsf{Aut}(\mathcal{R}_q) \cong \mathbb{Z}_{2\varphi}^\times$ to denote the automorphism $\sigma_{-1}(X) = X^{-1}$. For vectors, matrices, and tensors, $\sigma_{-1}(\cdot)$ applies the automorphism entry-wise. We also write $\bar{\cdot}$ to denote this automorphism operation for readability. We define a useful notation for the inverse coefficient embedding of 3-order tensors.

Definition 1 (Inverse Coefficient Embedding). *For a tensor $\mathcal{M} \in \mathbb{Z}^{c \times d \times \varphi}$, the inverse coefficient embedding cf^{-1} maps the third mode of dimension φ to the polynomial ring $\mathcal{R}$. It outputs a tensor $\mathsf{cf}^{-1}(\mathcal{M}) \in \mathcal{R}^{c \times d \times 1}$, with its $(i,j,1)$-th entry defined as*

$$\left(\mathsf{cf}^{-1}(\mathcal{M})\right)(i,j,1) := \sum_{\ell=0}^{\varphi-1} \mathcal{M}(i,j,\ell+1) X^\ell$$

for $i \in [c]$ and $j \in [d]$.

Norm. We define the norm of an element $x \in \mathcal{R}$ as the ℓ_2-norm of its coefficient embedding, denoted by $\|x\| := \|\mathsf{cf}(x)\|_2$. This definition extends to a vector $\mathbf{x} \in \mathcal{R}^m$ as $\|\mathbf{x}\| := \|\mathsf{cf}(\mathbf{x})\|_2$, and to a matrix $\mathbf{W} \in \mathcal{R}^{m \times r}$ as $\|\mathbf{W}\| := \|\mathsf{cf}(\mathsf{Vec}(\mathbf{W}))\|_2$.

2.1 Multilinear Extension and Sumchecks over Rings

Definition 2 (Multilinear Extension). *Let V be an $\mathcal{R}$-module. The multilinear extension of a function $f : \{0,1\}^\ell \to V$ is the unique ℓ-variate polynomial $\mathsf{mle}[f] : \mathcal{R}^\ell \to V$ of individual degree 1 satisfying $\mathsf{mle}[f](\mathbf{z}) = f(\mathbf{z})$ for all $\mathbf{z} \in \{0,1\}^\ell$.*

For $d = 2^\ell$, by identifying the index set $[d]$ with $\{0,1\}^\ell$, arrays indexed by d are extended coordinate-wise over $\mathcal{R}^\ell$. A vector $\mathbf{w} \in \mathcal{R}^d$, a matrix $\mathbf{W} \in \mathcal{R}^{d \times r}$, and a 3rd-order tensor $\mathcal{M} \in \mathcal{R}^{r_1 \times d \times r_2}$ correspond to functions mapping $\{0,1\}^\ell$ into the $\mathcal{R}$-modules $\mathcal{R}$, $\mathcal{R}^r$, and $\mathcal{R}^{r_1 \times r_2}$ respectively, yielding their coordinate-wise multilinear extensions $\mathsf{mle}[\mathbf{w}]$, $\mathsf{mle}[\mathbf{W}]$, and $\mathsf{mle}[\mathcal{M}]$.

We recall the basic sumcheck reduction of knowledge from [KLN+26], but we leave the input and output relation of the protocol implicit, as they will always be clear from the context.

Lemma 1 (Basic Sumcheck, adapted from [KLN+26]). *Let $\mathcal{R}$ be a non-trivial ring, N, M be $\mathcal{R}$-modules, $\mathcal{C} \subseteq \mathcal{R}$ be a subset with invertible difference. Let $g \in N[x_1, \dots, x_\ell]$ and $t \in M$. Let $\Phi : N \to M$ be an $\mathcal{C}$-linear map. There exists a reduction of knowledge Π^{sc} from a summation claim*

$$\Phi\left(\sum_{(x_1,\dots,x_\ell)\in\{0,1\}^\ell} g(x_1, \dots, x_\ell)\right) = t$$

to an evaluation claim $\Phi(g(\mathbf{r})) = v$ where $\mathbf{r} = (r_1, \dots, r_\ell) \in \mathcal{C}^\ell$ are the verifier's random challenges. The protocol is public-coin, has completeness error 0, has soundness error $\leqslant \frac{\ell \mathsf{deg}(g)}{|\mathcal{C}|}$, where $\mathsf{deg}(g)$ denotes the maximal individual degree. The communication is bounded by $\ell(\mathsf{deg}(g)+1)$ elements in M from the prover and ℓ challenges in $\mathcal{C}$ from the verifier.

Distributions. The distribution $\mathcal{D}_{\mathsf{ghl}}$ is defined as the $\mathsf{Bin}(2, \frac{1}{2}) - 1$, i.e., it is 0 with probability 1/2 and 1 (resp. -1) with probability 1/4.

Lemma 2. *For any fixed vector $\mathbf{x} \in [-q/2, q/2]^m$, if $\|\mathbf{x}\|_\infty \geqslant B$, for any $B < q/4$ then*

$$\Pr_{\mathbf{m} \leftarrow \mathcal{D}_{\mathsf{ghl}}^m}\left[|\langle \mathbf{m}, \mathbf{x}\rangle \bmod q| < \frac{B}{2}\right] \leqslant \frac{1}{2}.$$

Proof. Let i be an index of an entry in $\mathbf{x}$ with magnitude at least B. After choosing all but the i-th entry in $\mathbf{m}$, at most one of the three values $\{-1, 0, 1\}$ yields $|\langle \mathbf{m}, \mathbf{x}\rangle \bmod q| < B/2$.

2.2 Hardness Assumption

We state the formulation of the lattice hardness assumption used in this paper.

Definition 3 (Module-SIS (MSIS) Assumption). *Let λ be a security parameter and $\mathsf{params} = (\mathcal{R}, q, \kappa, m, B, \mathcal{D})$ be parameterized by λ, where $\mathcal{R}$ is a ring, $q \in \mathbb{N}$ is a modulus, $B > 0$ is a norm bound, and $\mathcal{D}$ is a probability distribution over $\mathcal{R}_q^{\kappa \times m}$. The $\mathsf{SIS}_{\mathsf{params}}$ assumption states that, for any PPT adversary $\mathcal{A}$, the advantage function satisfies*

$$\mathsf{Adv}^{\mathsf{sis}}_{\mathsf{params},\mathcal{A}}(\lambda) := \Pr\left[\begin{array}{l|l} \mathbf{Ax} = \mathbf{0} \pmod q & \mathbf{A} \leftarrow \mathcal{D} \\ 0 < \|\mathbf{x}\|_2 \leqslant B & \mathbf{x} \leftarrow \mathcal{A}(\mathbf{A}) \end{array}\right] \leqslant \mathsf{negl}(\lambda).$$

Definition 3 captures the standard Module-SIS assumption when $\mathcal{D}$ is the uniform distribution over $\mathcal{R}_q^{\kappa\times m}$. It captures the vanishing SIS (vSIS) assumption when $m = \prod_{i=1}^{\nu} m_i$ and $\mathcal{D}$ samples matrices exhibiting a row-tensor structure $\mathbf{A} = \mathbf{A}_1 \bullet \cdots \bullet \mathbf{A}_\nu$, where $\mathbf{A}_i \in \mathcal{R}_q^{\kappa\times m_i}$ for $i \in [\nu]$ and $\bullet$ stands for row-wise Kronecker product.

2.3 Relations

SIS relation. We define the relation Ξ^{sis} of the SIS problem:

$$\Xi^{\mathsf{sis}}_{\kappa,m,B} := \left\{ \begin{array}{l} (\mathbf{A};\mathbf{x}) : \\ \quad \mathbf{A} \in \mathcal{R}_q^{\kappa\times m},\ \mathbf{x} \in \mathcal{R}^m \\ \quad \mathbf{A}\mathbf{x} = \mathbf{t} \bmod q \ \wedge\ \|\mathbf{x}\|_2 \leqslant B \end{array} \right\}.$$

Polynomial Map and Sumcheck Relations. Let $B < q/2$ be the norm bound and $\mathbf{A} \in \mathcal{R}_q^{\kappa\times m}$ be an Ajtai commitment key. Let $f : \mathcal{R}^{mr} \to \mathcal{R}^n$ be a polynomial map. We define the polynomial map relation as follows:

$$\Xi^{\mathsf{poly}}_{f,\kappa,n,m,r,B,\varrho} := \left\{ \begin{array}{l} (\mathbf{T};\ (\mathbf{W},\mathbf{S})) : \\ \quad \mathbf{T} \in \mathcal{R}_q^{\kappa\times r},\ \mathbf{W} \in \mathcal{R}^{m\times r},\ \text{diagonal } \mathbf{S} \in \mathcal{R}^{r\times r}, \\ \quad \|\mathbf{S}\|_{\mathsf{op}} \leqslant \varrho \ \text{ s.t. } \ \|\mathbf{W}\mathbf{S}\|_2 < B, \\ \quad \mathbf{A}\mathbf{W} = \mathbf{T} \bmod q \ \wedge\ f(\mathbf{W}) = \mathbf{0} \bmod q \end{array} \right\}.$$

Let $m\cdot r = 2^\nu$ for $d \geqslant 2$. The sumcheck relation Ξ^{sum} augments Ξ^{poly} with n' additional sumcheck claims:

$$\Xi^{\mathsf{sum}}_{f,\kappa,n,n',m,r,B,\varrho} := \left\{ \begin{array}{l} (\mathbf{T};\ (\mathbf{W},\mathbf{S})) \in \Xi^{\mathsf{poly}}_{f,\kappa,n,m,r,B,\varrho} : \\ \quad \forall i \in [n'],\ \displaystyle\sum_{\mathbf{z}\in\{0,1\}^\nu} g_i\left(\mathsf{mle}[\mathsf{vec}(\mathbf{W})], \mathsf{vec}(\overline{\mathbf{W}})\right)(\mathbf{z}) = 0 \bmod q \end{array} \right\}.$$

We say that a relation is *relaxed* (denoted by $\Xi^{\mathsf{poly\text{-}rel}}$ or $\Xi^{\mathsf{sum\text{-}rel}}$) if the slack parameter satisfies $\varrho > 1$. If ϱ is omitted, it is implicitly set to 1, in which case we assume without loss of generality that $\mathbf{S} = \mathbf{I}_r$.

3 Tensor Train Random Projection

The *tensor train* (TT) *decomposition* [Ose11] is a generalization of the singular value decomposition that can represent an μ-th order tensor $\mathcal{M} \in \mathbb{R}^{d\times\cdots\times d}$ (or equivalently, a vector $\mathbf{m} \in \mathbb{R}^{d^\mu}$) using only $\mathcal{O}(c^2 d\mu)$ parameters, where the parameter c bounds the structural complexity of the decomposition. Informally, an arbitrary tensor $\mathcal{M} \in \mathbb{R}^{d\times\cdots\times d}$ of order μ is said to be represented in a TT-format with internal rank c if its individual entries can be exactly expressed as a chain of matrix multiplications

$$\forall (i_1,\ldots,i_\mu) \in [d]^\mu, \quad \mathcal{M}(i_1,i_2,\ldots,i_\mu) = \mathcal{M}_1(i_1)\mathcal{M}_2(i_2)\cdots\mathcal{M}_\mu(i_\mu), \tag{8}$$

where each $\mathcal{M}_n(i_n)$ is the i_n-th matrix slice of size $c_{n-1}\times c_n$ extracted from the n-th core tensor $\mathcal{M}_n \in \mathbb{R}^{c_{n-1}\times d\times c_n}, n \in [\mu]$. Since the output of a TT is a $c_0 \times c_\mu$ matrix, we fix $c_0 = c_\mu = 1$ so as to enforce the output to be a scalar, while the intermediate internal ranks are constant

$c_1 = \cdots = c_{\mu-1} = c$[8]. For algebraic convenience, we frequently employ an equivalent Kronecker product formulation. Under the same parameter definitions, the TT-format for a vectorized tensor $\mathbf{m} \in \mathbb{R}^{d^\mu}$ can be compactly given by

$$\mathbf{m} = \mathsf{Mat}\left(\mathcal{M}_1 \otimes \cdots \otimes \mathcal{M}_\mu\right).$$

Building upon this tensorized structure, the *tensor train random projection* (TTRP) [RR20] was introduced to efficiently reduce the dimensionality of high-order tensors isometrically. A TTRP $f_{\mathsf{TT}} : \mathbb{R}^{d^\mu} \to \mathbb{R}^k$ ($d^\mu \gg k$), is defined component-wise for the j-th output coordinate as

$$\left(f_{\mathsf{TT}}(\mathbf{x})\right)^{(j)} := \frac{1}{\sqrt{k}}\mathsf{Mat}\left(\mathcal{M}_1^{(j)} \otimes \cdots \otimes \mathcal{M}_\mu^{(j)}\right) \cdot \mathbf{x}, \quad \text{for } j \in [k], \tag{9}$$

where each core tensor $\mathcal{M}_i^{(j)}, i \in [\mu]$ is independently sampled with Gaussian[9] [RR20] or Rademacher [RR21] entries. The TTRP satisfies similar approximate-isometry properties as Johnson-Lindenstrauss projections. More precisely, it acts as an expected isometry that preserves the Euclidean geometry of the original high-dimensional space. Concretely, for any $\epsilon > 0$, the squared Euclidean norm of the projected vector is tightly preserved within a bounded multiplicative distortion, satisfying

$$\|f_{\mathsf{TT}}(\mathbf{x})\|_2^2 = (1 \pm \epsilon)\|\mathbf{x}\|_2^2$$

with a certain high probability.

3.1 Adaptation to Modular Arithmetic

The usual definition of TTRP [RR20] cannot be directly applied to lattice-based cryptography, as we work over discrete structures rather than the real numbers, and modulo a prime number q. To adapt TTRP to our setting, we introduce two main modifications:

1. We eliminate the fractional normalization coefficient $1/\sqrt{k}$ in Eq.(9).
2. The Gaussian (or Rademacher) distributions are replaced by $\mathcal{D}_{\mathsf{ghl}}$.

Definition 4 (Tensor Train Random Projection (TTRP), adapted from [RR20]). *Let $\mu, c, d, \varphi \geqslant 1$ and $m = d^\mu$ be integers. The tensor train random projection of internal rank $c \geqslant 1$ is a randomized linear map $f^k_{\mathsf{TT}(\mu,c)} : \mathbb{Z}^m \to \mathbb{Z}^k$ defined component-wise by*

$$\left(f^k_{\mathsf{TT}(\mu,c)}(\mathbf{x})\right)^{(j)} := \mathsf{Mat}\left(\mathcal{M}_1^{(j)} \otimes \cdots \otimes \mathcal{M}_\mu^{(j)}\right) \cdot \mathbf{x}, \quad \textit{for } j \in [k], \tag{10}$$

where $\mathcal{M}_1^{(j)} \in \mathbb{Z}^{1\times d\times c}$, $\mathcal{M}_i^{(j)} \in \mathbb{Z}^{c\times d\times c}$ for $1 < i < \mu$, and $\mathcal{M}_\mu^{(j)} \in \mathbb{Z}^{c\times d\times 1}$ are 3rd-order tensors with i.i.d. entries drawn from the distribution $\mathcal{D}_{\mathsf{ghl}}$. For the special case of $k = 1$, we omit the superscript and simply write $f_{\mathsf{TT}(\mu,c)}$ to denote the one-row tensor train random projection.

We also write $\{\mathcal{M}_i^{(j)}\}_{i\in[\mu],j\in[k]} \leftarrow \chi_{\mathsf{TT}}(k, m, d, \mu, c)$ to denote the sampling of the core tensors and

$$\mathsf{TT}(\{\mathcal{M}_i^{(j)}\}_{i\in[\mu],j\in[k]}) = \begin{bmatrix} \vdots \\ \mathsf{Mat}\left(\mathcal{M}_1^{(j)} \otimes \cdots \otimes \mathcal{M}_\mu^{(j)}\right) \\ \vdots \end{bmatrix} \in \mathbb{Z}^{k\times m}$$

to denote the resulting k-row TTRP matrix for simplicity.

[8] The general definition of the TT-decomposition allows the dimension d and internal rank c to vary across core tensors, but this uniform formulation is sufficient for this paper.

[9] Specifically, $\mathcal{N}(0, c^{-1/2})$ for $i \in \{1, \mu\}$ and $\mathcal{N}(0, c^{-1})$ for $1 < i < \mu$.

Lemma 3 (Second and Fourth Moments). *Let $\mathbf{x} \in \mathbb{Z}^m$ be a fixed vector. For the one-row tensor train random projection $y = f_{\mathsf{TT}(\mu,c)}(\mathbf{x})$, we have*

$$\mathbb{E}\left[y^2\right] = \frac{c^{\mu-1}}{2^\mu}\|\mathbf{x}\|_2^2, \text{ and} \tag{11}$$

$$\mathbb{E}\left[y^4\right] \leqslant \frac{3}{4}\left(\frac{c(c+2)}{4}\right)^{\mu-1}\|\mathbf{x}\|_2^4. \tag{12}$$

The proof is deferred to Section A.2

3.2 Bounding Modular Overflows

Lemma 4. *Let $\mathbf{x} \in [-q/2, q/2]^m$ be a fixed vector such that $\|\mathbf{x}\|_\infty \geqslant B$ for some $B < q/4$. For the one-row tensor train random projection $f_{\mathsf{TT}(\mu,c)}$, we have*

$$\Pr\left[\left|f_{\mathsf{TT}(\mu,c)}(\mathbf{x}) \bmod q\right| < \frac{B}{2^\mu}\right] \leqslant \frac{1}{2} + \frac{\mu-1}{2^{c+1}}. \tag{13}$$

Proof. Let $y := f_{\mathsf{TT}(\mu,c)}(\mathbf{x}) \bmod q = \mathbf{p}^\top\mathbf{x} \bmod q$, where $\mathbf{p}^\top := \mathsf{Mat}\left(\mathcal{M}_1 \otimes \cdots \otimes \mathcal{M}_\mu\right) \in \mathbb{Z}^{1\times m}$. By applying the mixed-product property (Fact 1), we evaluate this inner product iteratively from right to left. Defining $\mathbf{M}_i := \mathsf{Mat}(\mathcal{M}_i)$, we construct the sequence of intermediate computation state $\mathbf{v}^{(i)}$:

$$\begin{aligned}
\mathbf{v}^{(\mu)} &:= \mathbf{x}, \\
\mathbf{v}^{(i-1)} &:= \left(\mathbf{I}_{d^{i-1}} \otimes \mathbf{M}_i\right)\mathbf{v}^{(i)} \bmod q, \quad \text{for } i = \mu, \mu-1, \ldots, 2, \\
\mathbf{v}^{(0)} &:= \mathbf{M}_1\mathbf{v}^{(1)} \bmod q = y.
\end{aligned}$$

Let E_i denote the event that $\|\mathbf{v}^{(i)}\|_\infty \geqslant B/2^{\mu-i}$. By hypothesis, $\|\mathbf{v}^{(\mu)}\|_\infty = \|\mathbf{x}\|_\infty \geqslant B$, yielding $\Pr[E_\mu] = 1$. For any $2 \leqslant i \leqslant \mu$, assume E_i holds. We partition the column vector $\mathbf{v}^{(i)}$ into d^{i-1} blocks. Since $\|\mathbf{v}^{(i)}\|_\infty \geqslant B/2^{\mu-i}$, there exists at least one block $\mathbf{u}^*$ satisfying $\|\mathbf{u}^*\|_\infty \geqslant B/2^{\mu-i}$. Applying Lemma 2 to each row of $\mathbf{M}_i$, the probability that the absolute value of a single component of $\mathbf{M}_i\mathbf{u}^*$ is less than $B/2^{\mu-i+1}$ is bounded by $1/2$. Since the c rows are sampled independently, the probability that at least one component preserves the bound $B/2^{\mu-i+1}$ satisfies $\Pr[E_{i-1} \mid E_i] \geqslant 1 - 2^{-c}$. For the final projection $y = \mathbf{M}_1\mathbf{v}^{(1)} \bmod q$, $\mathbf{M}_1 \in \mathbb{Z}^{1\times cd}$ consists of a single row. Conditioning on E_1, Lemma 2 yields $\Pr[E_0 \mid E_1] \geqslant 1/2$. The probability of E_0 (the event $|y| \geqslant B/2^\mu$) is bounded using Bernoulli's inequality

$$\Pr[E_0] \geqslant \frac{1}{2}\left(1 - 2^{-c}\right)^{\mu-1} \geqslant \frac{1}{2}\left(1 - \frac{\mu-1}{2^c}\right) = \frac{1}{2} - \frac{\mu-1}{2^{c+1}}. \tag{14}$$

Therefore, $\Pr[\neg E_0] = 1 - \Pr[E_0] \leqslant \frac{1}{2} + \frac{\mu-1}{2^{c+1}}$.

Theorem 2. *Let $\mu, c, k \geqslant 1$. For any fixed vector $\mathbf{x} \in [-q/2, q/2]^m$ satisfying $\|\mathbf{x}\|_\infty \geqslant B_\infty$ for some $B_\infty < q/4$, the k-row projection $\mathbf{y} = f^k_{\mathsf{TT}(\mu,c)}(\mathbf{x}) \bmod q \in \mathbb{Z}_q^k$ satisfies*

$$\Pr\left[\|\mathbf{y}\|_\infty < \frac{B_\infty}{2^\mu}\right] \leqslant \left(\frac{1}{2} + \frac{\mu-1}{2^{c+1}}\right)^k.$$

Furthermore, if $\|\mathbf{x}\|_2 \geqslant B$ for some $B < q\sqrt{m}/4$, the projection satisfies

$$\Pr\left[\|\mathbf{y}\|_2 < \frac{B}{\sqrt{m}2^\mu}\right] \leqslant \left(\frac{1}{2} + \frac{\mu-1}{2^{c+1}}\right)^k.$$

For any constant integer $\delta \geqslant 1$, setting the internal rank $c \geqslant \lceil \log_2(\mu-1)\rceil + \delta$ and the row count $k \geqslant \frac{\lambda}{-\log_2(1/2+2^{-\delta-1})} \in \mathcal{O}(\lambda)$ bounds this probability by $2^{-\lambda}$.

Proof. We first prove the bound for the ℓ_∞ norm. The mapping generates k independent outputs $y_i = \left(f^k_{\mathsf{TT}(\mu,c)}(\mathbf{x})\right)^{(i)} \bmod q$. By Lemma 4, the probability that a single component is bounded by the threshold is

$$p_{\mathsf{fail}} := \Pr\left[|y_i| < \frac{B_\infty}{2^\mu}\right] \leqslant \frac{1}{2} + \frac{\mu-1}{2^{c+1}}.$$

The condition $\|\mathbf{y}\|_\infty < B_\infty/2^\mu$ necessitates $|y_i| < B_\infty/2^\mu$ for all $i \in [k]$. Since the k rows are sampled independently, the probability of this joint event is bounded by $(p_{\mathsf{fail}})^k$

$$\Pr\left[\|\mathbf{y}\|_\infty < \frac{B_\infty}{2^\mu}\right] \leqslant (p_{\mathsf{fail}})^k \leqslant \left(\frac{1}{2} + \frac{\mu-1}{2^{c+1}}\right)^k.$$

The bound for the ℓ_2 norm follows from algebraic relations. The hypothesis $\|\mathbf{x}\|_2 \geqslant B$ implies $\|\mathbf{x}\|_\infty \geqslant B/\sqrt{m}$. Substituting $B_\infty = B/\sqrt{m}$ into the ℓ_∞ bound yields $\Pr\left[\|\mathbf{y}\|_\infty < \frac{B}{\sqrt{m}2^\mu}\right] \leqslant (p_{\mathsf{fail}})^k$. Since $\|\mathbf{y}\|_2 \geqslant \|\mathbf{y}\|_\infty$, the event $\|\mathbf{y}\|_2 < \frac{B}{\sqrt{m}2^\mu}$ implies $\|\mathbf{y}\|_\infty < \frac{B}{\sqrt{m}2^\mu}$. Thus, the probability is bounded by the same value.

Substituting $c \geqslant \lceil \log_2(\mu-1) \rceil + \delta$ yields $2^{c+1} \geqslant (\mu-1)2^{\delta+1}$. This bounds the per-row failure probability by $p_{\mathsf{fail}} \leqslant 1/2 + 2^{-\delta-1}$. To bound the overall probability by $2^{-\lambda}$ requires $(1/2+2^{-\delta-1})^k \leqslant 2^{-\lambda}$, which yields the sufficient condition $k \geqslant \lambda/(-\log_2(1/2+2^{-\delta-1}))$.

3.3 ℓ_2 Concentration Bounds and Slack Factor.

Theorem 2 provides a probabilistic bound for the ℓ_∞ norm mod q. A key advantage of this ℓ_∞-based analysis is its parameter efficiency. Instantiating the TTRP with $d \in \mathcal{O}(1)$ and $c \in \mathcal{O}(\log \mu)$ achieves λ bits of soundness for a projection dimension $k \in \mathcal{O}(\lambda)$, yielding the asymptotic representation size of $\mathcal{O}(\lambda \log m (\log\log m)^2) \in \tilde{\mathcal{O}}(\lambda)$.

Applying norm equivalence bounds on Theorem 2 to obtain an ℓ_2-concentration bound incurs a large factor $\frac{1}{\sqrt{m}2^\mu}$. To achieve a tighter ℓ_2 slack, we introduce a variance-based analysis in Lemma 5 and Theorem 3. Compared to Theorem 2, our ℓ_2 slack factor is tighter, while the internal rank increases to $c = \mathcal{O}(\log m)$. We note that the overall complexity remains $\tilde{\mathcal{O}}(\lambda)$.

Lemma 5. *Let $\mathbf{x} \in \mathbb{Z}^m$ be fixed, $\eta_2 = \frac{c^{\mu-1}}{2^\mu}$, and $\eta_4 = \frac{3}{4}\left(\frac{c(c+2)}{4}\right)^{\mu-1}$. For the one-row TTRP $y = f_{\mathsf{TT}(\mu,c)}(\mathbf{x})$ and any $\theta \in (0, \eta_2)$, we have*

$$\Pr\left[y^2 > \theta\|\mathbf{x}\|_2^2\right] \geqslant \frac{(\eta_2-\theta)^2}{\eta_4 - 2\eta_2\theta + \theta^2}.$$

For the k-row projection $\mathbf{y} = f^k_{\mathsf{TT}(\mu,c)}(\mathbf{x})$, we have

$$\Pr\left[\|\mathbf{y}\|_2 \leqslant \sqrt{\theta}\|\mathbf{x}\|_2\right] \leqslant \left(1 - \frac{(\eta_2-\theta)^2}{\eta_4 - 2\eta_2\theta + \theta^2}\right)^k. \tag{15}$$

Proof. By Lemma 3, the moments of y^2 are $\mathbb{E}[y^2] = \eta_2\|\mathbf{x}\|_2^2$ and $\mathbb{E}[y^4] \leqslant \eta_4\|\mathbf{x}\|_2^4$. Cantelli's inequality states that for any random variable Z with expectation μ_Z and variance σ_Z^2, and for any $t > 0$: $\Pr[Z > \mu_Z - t] \geqslant \frac{t^2}{\sigma_Z^2 + t^2}$.

Let $Z = y^2$ with $\mu_Z = \mathbb{E}[y^2]$ and $\sigma_Z^2 = \mathbb{E}[y^4] - (\mathbb{E}[y^2])^2$. Setting the deviation $t = \mathbb{E}[y^2] - \theta\|\mathbf{x}\|_2^2 = (\eta_2 - \theta)\|\mathbf{x}\|_2^2 > 0$, we obtain

$$\Pr\left[y^2 > \theta\|\mathbf{x}\|_2^2\right] \geqslant \frac{t^2}{\mathbb{E}[y^4] - (\mathbb{E}[y^2])^2 + t^2} = \frac{(\eta_2-\theta)^2\|\mathbf{x}\|_2^4}{\mathbb{E}[y^4] - (2\eta_2\theta - \theta^2)\|\mathbf{x}\|_2^4}.$$

Substituting the upper bound $\mathbb{E}[y^4] \leqslant \eta_4 \|\mathbf{x}\|_2^4$ yields the single-row bound. Amplifying by k-row repetition, we get the Eq.(15).

Theorem 3. *Let $\mu, c, k \geqslant 1$. Let θ, η_2, η_4 be defined as in Lemma 5. Let $\mathbf{x} \in [-q/2, q/2]^m$ be a fixed vector such that $\|\mathbf{x}\|_2 \geqslant B$ for some $B < \frac{q}{2^{\mu+1}\sqrt{\theta} c^{\mu-1} m}$. Let $p := \max\left\{\left(1 - \frac{(\eta_2-\theta)^2}{\eta_4 - 2\eta_2\theta + \theta^2}\right)^k, \left(\frac{1}{2} + \frac{\mu-1}{2^{c+1}}\right)^k\right\}$. For the k-row projection $\mathbf{y} = f^k_{\mathsf{TT}(\mu,c)}(\mathbf{x}) \bmod q \in \mathbb{Z}_q^k$, we have*

$$\Pr\left[\|\mathbf{y}\|_2 \leqslant \sqrt{\theta} B\right] \leqslant p$$

Proof. We have two cases:

- If $\|\mathbf{x}\|_\infty \geqslant \frac{q}{2 \cdot c^{\mu-1} m}$, by Lemma 4, the probability that $\|\mathbf{y}\|_\infty$ smaller than $\frac{q}{2^{\mu+1} \cdot c^{\mu-1} m}$ is at most $\left(\frac{1}{2} + \frac{\mu-1}{2^{c+1}}\right)^k$, since $B < \frac{q}{2^{\mu+1}\sqrt{\theta} c^{\mu-1} m}$ then

$$\Pr\left[\|\mathbf{y}\|_2 \leqslant \sqrt{\theta} B\right] \leqslant \Pr\left[\|\mathbf{y}\|_2 < \frac{q}{2^{\mu+1} \cdot c^{\mu-1} m}\right] \leqslant \left(\frac{1}{2} + \frac{\mu-1}{2^{c+1}}\right)^k$$

- If $\|\mathbf{x}\|_\infty < \frac{q}{2 \cdot c^{\mu-1} m}$, here with probability one we have $f^k_{\mathsf{TT}(\mu,c)}(\mathbf{x}) \in [\pm q/2]^k$. So mod-$q$ reduction has no effect and the assertion follows directly from Lemma 5.

4 Shortness Proof via TTRP

In this section, we demonstrate how the tensor train random projection serves as a modular component to integrate approximate range proofs into reductions of knowledge for lattice relations.

4.1 Norm Check for Ξ^{poly}

We describe a norm check protocol for the Ξ^{poly} relation which is formalized as an interactive reduction of knowledge Π^0 (Figure 1) from the polynomial map relation Ξ^{poly} to the sumcheck relation Ξ^{sum}.

The core mechanism is to replace the norm bound in the relation Ξ^{poly} by tensor train random projection. The verifier sends the core tensors $\{\mathcal{M}_i^{(j)}\}_{i\in[\mu], j\in[k]}$ defining a TTRP $f^k_{\mathsf{TT}(\mu,c)}(\cdot)$, then the prover sends the projection of the witness coefficients $\mathbf{y}_0 = f^k_{\mathsf{TT}(\mu,c)}(\mathsf{cf}(\mathsf{vec}(\mathbf{W})))$. The verifier checks the expected bound $\|\mathbf{y}_0\|_2 \leqslant \hat{B} := \sqrt{k(c/2)^{\mu-1}} \cdot B$ which holds with probability at least $1/2$ and implies $\|\mathbf{W}\| \leqslant \hat{B}/\sqrt{\theta}$ with overwhelming probability p. To prove well-formedness, the projection is lifted to $\mathcal{R}_q$ and formulated as sumcheck claims over the ring. We note that TTRP can actually be captured by relation Ξ^{poly}, so it can also be regarded as a self-RoK.

Theorem 4. *Let $\nu, \mu, d, c, k, \theta, p$ be the TTRP parameters satisfying $mr = 2^\nu$ and $mr\varphi = d^\mu$. Let B be the norm bound defined as in Theorem 3. Let $\tilde{B} := \sqrt{\frac{kc^{\mu-1}}{\theta 2^{\mu-1}}} B$.*

- *Π^0 is correct with a correctness error of $1/2$, reducing the relation*

$$\Xi^{\mathsf{poly}}_n \to \Xi^{\mathsf{sum}}_{n,k'}.$$

- *Π^0 is knowledge sound with a knowledge error of $\delta_0 = p + q^{-k'}$ for the reduction*

$$\Xi^{\mathsf{poly}}_{\tilde{B}} \cup \Xi^{\mathsf{sis}}_{2B'\varrho} \leftarrow \Xi^{\mathsf{sum\text{-}rel}}_{B',\varrho}.$$

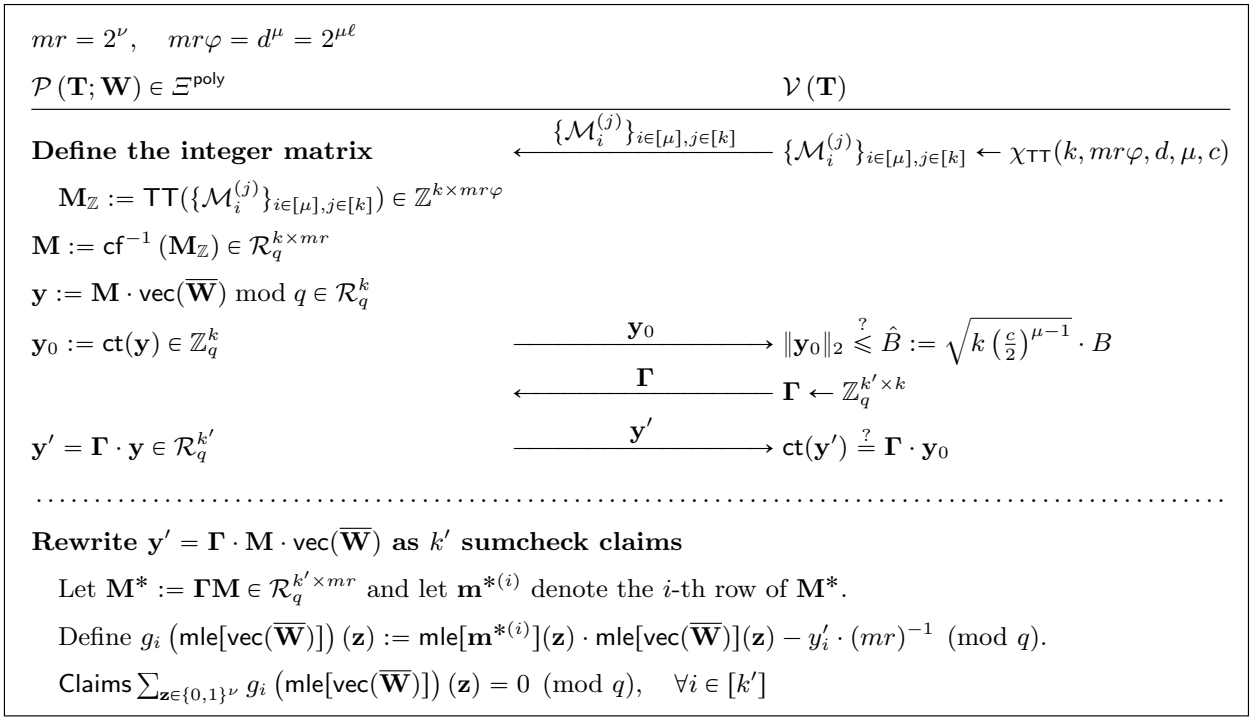


Fig. 1. Protocol Π^0 — RoK from Ξ^{poly} to Ξ^{sum}.

The proof is deferred to Section A.1

Remark 1 (Achieving $\mathcal{O}(\lambda)$ Communication Complexity). To achieve a soundness error of $2^{-\lambda}$ in Figure 1, the batching of k constraints over $\mathbb{Z}_q$ via the challenge matrix $\mathbf{\Gamma}$ requires a row dimension $k' = \Omega(\lambda/\log q)$. Transmitting the k' ring elements of the aggregated vector $\mathbf{y}'$ therefore yields a communication complexity at least quadratic in λ. For the purpose of asymptotic efficiency, this complexity can be reduced to $\tilde{\mathcal{O}}(\lambda)$ by applying the "Lift-and-Batch" technique from [KLNO25,KLN+26], which lifts multiple partial trace equalities to ring equalities via intermediate subrings.

Remark 2 (Variance-Based Completeness Bound). In the proof of Theorem 4, we establish the completeness bound $\hat{B}$ using Markov's inequality based on the expected squared norm. One can derive a variance-based upper bound using Cantelli's inequality on the sum of independent projection rows to parameterize the completeness error. While this approach tightens the constant factors of the verification check, it provides no asymptotic reduction to the extraction slack factor $\tilde{B}/B$. The slack is dominated by the divergence between the second and fourth moments of the TTRP distribution, which scales exponentially with the tensor order μ.

4.2 Linearisation via Sumcheck

Observe the k' sumcheck claims in the output relation Ξ^{sum} of Π^0 in Figure 1

$$\sum_{\mathbf{z}\in\{0,1\}^\nu} g_i(\mathbf{z}) = 0 \pmod q, \quad \forall i \in [k'],$$

where $g_i(\mathbf{z}) := \mathsf{mle}[\mathbf{m}^{*(i)}](\mathbf{z}) \cdot \mathsf{mle}[\mathsf{vec}(\overline{\mathbf{W}})](\mathbf{z}) - y_i' \cdot 2^{-\nu} \pmod q$. To batch these constraints, the verifier samples a random combiner $\boldsymbol{\gamma} = (\gamma_1, \ldots, \gamma_{k'}) \leftarrow \mathcal{R}_q^{k'}$ and sends it to the prover. We define

a batched summation claim as

$$\sum_{i=1}^{k'} \gamma_i \sum_{\mathbf{z}\in\{0,1\}^{\nu}} g_i(\mathbf{z}) = 0 \pmod{q}. \tag{16}$$

By invoking the basic sumcheck protocol (Lemma 1), the original k' summation claims are reduced to a single evaluation claim at a random challenge point $\mathbf{r} \leftarrow \mathcal{C}^{\nu}$ generated by the verifier

$$\sum_{i=1}^{k'} \gamma_i g_i(\mathbf{r}) = v \pmod{q},$$

where v is the terminal value claimed by the prover. To check this terminal relation, the verifier evaluates $\mathsf{mle}[\mathbf{m}^{*(i)}](\mathbf{r})$ locally. The prover then sends the evaluation $w_{\mathbf{r}} := \mathsf{mle}[\mathsf{vec}(\overline{\mathbf{W}})](\mathbf{r})$, allowing the verifier to check

$$v \stackrel{?}{=} \sum_{i=1}^{k'} \gamma_i \left(\mathsf{mle}[\mathbf{m}^{*(i)}](\mathbf{r}) \cdot w_{\mathbf{r}} - y'_i \cdot (mr))^{-1}\right) \pmod{q}. \tag{17}$$

The verifier then evaluates $\mathsf{mle}[\mathbf{m}^{*(i)}](\mathbf{r})$ locally. The prover sends the evaluation $w_{\mathbf{r}} := \mathsf{mle}[\mathsf{vec}(\overline{\mathbf{W}})](\mathbf{r})$, allowing the verifier to check

$$v \stackrel{?}{=} \sum_{i=1}^{k'} \gamma_i \left(\mathsf{mle}[\mathbf{m}^{*(i)}](\mathbf{r}) \cdot w_{\mathbf{r}} - y'_i \cdot (mr))^{-1}\right) \pmod{q}. \tag{18}$$

Since conjugation over $\mathcal{R}_q$ is an automorphism, it satisfies the identity

$$\mathsf{mle}[\mathsf{vec}(\overline{\mathbf{W}})](\mathbf{r}) = \overline{\mathsf{mle}[\mathsf{vec}(\mathbf{W})](\bar{\mathbf{r}})}$$

for any challenge point $\mathbf{r} \in \mathcal{R}_q^{\nu}$. The verifier conjugates the prover's message to obtain $\mathsf{mle}[\mathsf{vec}(\mathbf{W})](\bar{\mathbf{r}}) = \overline{w_{\mathbf{r}}}$, and appends this evaluation claim on the original vectorized witness at the conjugated point $\bar{\mathbf{r}}$ to the relation Ξ^{poly} as a new polynomial map constraint. From now on, we abuse the notation $\mathsf{mle}[\mathsf{vec}(\mathbf{W})](\mathbf{r}) = w_{\mathbf{r}}$ to denote this evaluation claim, given this equivalence.

Corollary 1. *Let $\mathcal{R}_q \cong \mathbb{F}_{q^e}{}^{\varphi/e}$. Let $\mathcal{C}$ be the challenge set for the sumcheck protocol and δ_0 be the knowledge error of Π^0 (Figure 1). Assume a TTRP instantiated with the parameters from Theorem 4. Let Π^{sc} be basic sumcheck protocol (Lemma 1). There exists a self-RoK Π^{TTRP} for the relation Ξ^{poly} defined as the composition $\Pi^{\mathsf{TTRP}} = \Pi^{\mathsf{sc}} \circ \Pi^0$ such that*

- *Π^{TTRP} is complete with a completeness error of $1/2$, reducing the relation*

$$\Xi_n^{\mathsf{poly}} \to \Xi_{n+1}^{\mathsf{poly}}.$$

- *Π^{TTRP} is knowledge sound with a knowledge error of $\delta_0 + q^{-e} + \frac{2\nu}{|\mathcal{C}|}$ for the reduction*

$$\Xi_{\tilde{B}}^{\mathsf{poly}} \cup \Xi_{2B'\varrho}^{\mathsf{sis}} \leftarrow \Xi_{B',\varrho}^{\mathsf{poly\text{-}rel}},$$

  where $\tilde{B}$ is defined as in Theorem 4.

Proof. The completeness error follows from Lemma 1 and Theorem 4. The knowledge error is bounded by a union bound over three terms

- δ_0: The knowledge error of Π^{0}.
- q^{-e}: The collision probability of the random linear combination in Eq.(16). Since $\boldsymbol{\gamma} \leftarrow \mathcal{R}_q^{k'}$ and $\mathcal{R}_q \cong \mathbb{F}_{q^e}{}^{\varphi/e}$, a non-zero vector vanishes with probability at most q^{-e}.
- $\frac{2\nu}{|\mathcal{C}|}$: The soundness error of the basic sumcheck protocol (Lemma 1) for ν variables, as the maximum individual degree of g_i is 2.

Lemma 6 (Efficiency of Π^{TTRP}). *Let $\mathcal{C}$ be the sumcheck challenge set. The protocol Π^{TTRP} (Corollary 1) has the following costs, where $mr = 2^\nu$ and $mr\varphi = d^\mu = 2^{\mu\ell}$*

- *Prover runtime: $\mathcal{O}(kcmr\varphi)$ operations over $\mathbb{Z}_q$ and $\mathcal{O}(mr)$ operations over $\mathcal{R}_q$. This includes evaluating the TTRP projection, batching k' sumcheck claims, and executing the ν-variate basic sumcheck for a degree 2 polynomial.*
- *Verifier runtime Sampling $\mathcal{O}(kc^2d\mu)$ elements from $\mathcal{D}_{\mathsf{ghl}}$ and $\mathcal{O}(k'k)$ elements from $\mathbb{Z}_q$. Evaluating the ν-variate multilinear extension via block-wise tensor contraction requires $\mathcal{O}(k\mu_1 c^2 d)$ operations over $\mathcal{R}_q$ and $\mathcal{O}(kc\varphi)$ additions. Forming the aggregated claims requires $\mathcal{O}(k'k)$ operations over $\mathcal{R}_q$.*
- *Prover-to-verifier communication: k elements over $\mathbb{Z}_q$, $k'+1$ elements over $\mathcal{R}_q$ and 2ν elements over $\mathbb{F}_{q^e}$.*

Proof. *Prover Runtime* The prover runtime is dominated by the computation of the projection $\mathbf{y}_0 = \mathbf{M}_{\mathbb{Z}} \cdot \mathsf{cf}(\mathsf{vec}(\mathbf{W})) \pmod q$. We focus on this computation, other costs follow directly from the protocol description. The projection is evaluated via right-to-left tensor contraction which takes the flattened coefficient vector $\mathbf{x} := \mathsf{cf}(\mathsf{vec}(\mathbf{W})) \in \mathbb{Z}_q^{d^\mu}$ as input. The contraction sequence computes $\mathbf{v}_{i-1} = \left(\mathbf{I}_{d^{i-1}} \otimes \mathbf{M}_i^{(j)}\right) \mathbf{v}_i \pmod q$ for $i = \mu, \mu-1, \ldots, 1$, where $\mathbf{M}_i^{(j)} := \mathsf{Mat}\left(\mathcal{M}_i^{(j)}\right)$. Since the internal matrices $\mathbf{M}_i^{(j)}$ are over $\mathbb{Z}$, the sequence consists exclusively of integer multiplications. The total cost for k rows is dominated by the initial layers, yielding $\mathcal{O}(kcd^\mu) = \mathcal{O}(kcmr\varphi)$ operations over $\mathbb{Z}_q$.

Verifier Runtime We analyze the computational cost for the verifier to evaluate the multilinear extension $\mathsf{mle}[\mathbf{m}^{*(i)}](\mathbf{r})$ for the i-th aggregated row, where $i \in [k']$. By the linearity of the multilinear extension, this evaluation can be expressed as $\mathsf{mle}[\mathbf{m}^{*(i)}](\mathbf{r}) = \boldsymbol{\Gamma}^{(i)}\mathsf{mle}[\mathbf{M}](\mathbf{r})$. Here, $\boldsymbol{\Gamma}^{(i)}$ is the i-th row of the challenge matrix $\boldsymbol{\Gamma}$, and $\mathsf{mle}[\mathbf{M}](\mathbf{r})$ is a column vector containing the evaluations $\mathsf{mle}[\mathbf{M}^{(j)}](\mathbf{r})$ for each original row $j \in [k]$.

Let $mr = 2^\nu$ be the spatial dimension. The verifier computes $\mathsf{mle}[\mathbf{M}^{(j)}](\mathbf{r})$ directly without explicitly computing the length-mr polynomial vector as follows. The i-th polynomial in the vector $\mathbf{M}^{(j)}$ can be expanded as $\mathbf{M}_i^{(j)} = \sum_{h=0}^{\varphi-1} \mathbf{M}_{\mathbb{Z},i,h}^{(j)} X^h$. In this expansion, X^h is the polynomial basis, and $\mathbf{M}_{\mathbb{Z},i,h}^{(j)}$ is the integer entry of the underlying tensor network at spatial index $i \in \{0,1\}^\nu$ and coefficient index $h \in \{0, \ldots, \varphi-1\}$, i.e., i-th polynomial's h-th coefficient.

Let $\chi_i(\mathbf{r})$ denote the multilinear basis polynomial evaluated at the challenge point $\mathbf{r} \in \mathcal{C}^\nu$. The evaluation of the polynomial vector expands and rearranges as

$$\mathsf{mle}[\mathbf{M}^{(j)}](\mathbf{r}) = \sum_{i\in\{0,1\}^\nu} \left(\sum_{h=0}^{\varphi-1} \mathbf{M}_{\mathbb{Z},i,h}^{(j)} X^h\right) \chi_i(\mathbf{r}) = \sum_{h=0}^{\varphi-1} \left(\sum_{i\in\{0,1\}^\nu} \mathbf{M}_{\mathbb{Z},i,h}^{(j)} \chi_i(\mathbf{r})\right) X^h$$

This equation shows that the h-th coefficient of the evaluation $\mathsf{mle}[\mathbf{M}^{(j)}](\mathbf{r})$ is exactly the spatial multilinear extension of the h-th coefficient slice of the tensor, evaluated at $\mathbf{r}$.

The verifier computes these φ coefficients simultaneously by exploiting the tensor structure. Let $d = 2^\ell$ be the core dimension. The μ tensor cores partition into $\mu_1 = \nu/\ell$ spatial cores and $\mu_2 = (\log_2 \varphi)/\ell$ coefficient cores. Let c be the internal rank.

The spatial multilinear extension is independent of the coefficient index h. The verifier chunks the challenge point $\mathbf{r}$ into μ_1 blocks of length ℓ. Evaluating the multilinear extension of each spatial core at the corresponding chunk and computing their chain matrix product yields a $1 \times c$ boundary vector $\mathbf{v}^{(j)}(\mathbf{r})$. This step contracts the spatial dimensions and requires $\mathcal{O}(\mu_1 c^2 d)$ operations over $\mathcal{R}_q$.

To recover the φ evaluated coefficients, the verifier multiplies the boundary vector $\mathbf{v}^{(j)}(\mathbf{r})$ by the remaining μ_2 coefficient cores. Since these coefficient cores contain only elements from $\{-1, 0, 1\}$, extracting all φ coefficients requires $\mathcal{O}(c\varphi)$ additions. Applying the embedding maps these coefficients to the final evaluated ring element in $\mathcal{R}_q$.

For all k rows, this tensor-structured coefficient evaluation takes $\mathcal{O}(k(\mu_1 c^2 d + c\varphi))$ operations. Finally, computing the k' linear combinations with the challenge matrix $\mathbf{\Gamma}$ takes $\mathcal{O}(k'k)$ operations over $\mathcal{R}_q$.

5 Integration to Proof Systems

In this section, we instantiate the integration of our TTRP shortness proof modules Π^{0} and Π^{TTRP} into the lattice-based proof systems LaBRADOR [BS23] and `RoKoko` [KLN+26]. We assume familiarity with these protocols and recall their shortness proof mechanisms.

5.1 Shortness Proofs in LaBRADOR and `RoKoko`

To demonstrate that a committed witness $\mathbf{W} := (\mathbf{w}_1, \ldots, \mathbf{w}_r) \in \mathcal{R}^{m \times r}$ satisfies the norm bound $\|\mathbf{W}\| \leqslant B$, LaBRADOR applies the modular Johnson-Lindenstrauss lemma. The lemma establishes that for some $\beta, p_\beta > 0$ and every $\mathbf{x} \in \mathbb{Z}^D$, the projection satisfies

$$\Pr_{\mathbf{M} \leftarrow \mathcal{D}_{\mathsf{ghl}}^{256 \times D}} [\|\mathbf{M}\mathbf{x}\| > \beta \|\mathbf{x}\|] \lesssim p_\beta{}^{10}.$$

For every vector $\mathbf{x} \in [\pm q/2]^D$ satisfying $\|\mathbf{x}\| \geqslant B$ for a bound $B \leqslant q/125$, the probability bound is given by

$$\Pr_{\mathbf{M} \leftarrow \mathcal{D}_{\mathsf{ghl}}^{256 \times D}} \left[\|\mathbf{M}\mathbf{x} \bmod q\| < \sqrt{30} B \right] \lesssim 2^{-128}.$$

The verifier samples a projection matrix $\mathbf{M} \leftarrow \mathcal{D}_{\mathsf{ghl}}^{256 \times mr\varphi}$. The prover computes the projected vector $\mathbf{y}_0 := \mathbf{M} \cdot \mathsf{cf}(\mathsf{vec}(\mathbf{W})) \bmod q \in \mathbb{Z}_q^{256}$. The verifier checks whether $\|\mathbf{y}_0\| \leqslant \beta B$. This condition holds with probability $1 - p_\beta$ and implies $\|\mathbf{W}\| < \beta B / \sqrt{30}$ with probability $1 - 2^{-128}$, introducing a slack factor of $\beta/\sqrt{30}$. To prove the well-formedness of $\mathbf{y}_0$, the projection is expressed and aggregated as a linear relation over $\mathcal{R}_q$, compatible with the relation of the protocol. Processing the projection matrix $\mathbf{M}$ requires $\mathcal{O}(\lambda mr)$ verifier runtime.

The shortness proof paradigm in `RoKoko` follows an analogous approach while preserving succinct verification. We outline the differences below.

– *Structured Projection.* `RoKoko` adopts the block-diagonal structured random projection from RoK and Roll [KLNO25]. For the *coarse projection*, the operation over the ring $\mathcal{R}_q$ is defined as

$$\mathbf{V} = \left(\mathbf{I}_{m/m_{\mathrm{rp}}} \otimes \mathbf{M} \right) \cdot \mathbf{W}$$

[10] The expansion factor β determines the trade-off between the extraction slack and the prover's abort rate. LaBRADOR[BS23] instantiates β to match the expected squared norm. The completeness error p_β is bounded inherently by the right-skewness of the underlying (heuristic) chi-squared distribution of the squared projected norm, which ensures sufficient probability mass resides below the expectation. The prover satisfies this condition via rejection sampling. Alternatively, selecting a larger β yields an abort-free prover at the cost of a looser bound.

where $\mathbf{M} \leftarrow \mathcal{D}_{\mathsf{ghl}}^{n_{\mathtt{rp}} \times m_{\mathtt{rp}}}$. The prover applies a gadget decomposition $\mathbf{Y}_{\mathsf{proj}} = \mathbf{G}^{-1}(\mathbf{V})$ and commits to $\mathsf{vec}(\mathbf{Y}_{\mathsf{proj}})$, establishing the updated constraint $(\mathbf{I}_{m/m_{\mathtt{rp}}} \otimes \mathbf{M})\mathbf{W} = \mathbf{G}\mathbf{Y}_{\mathsf{proj}} \pmod q$. The *fine projection* over the $\mathbb{Z}_q$ coefficient embedding is evaluated as

$$\mathbf{V} = (\mathbf{I}_{\varphi m/m_{\mathtt{rp}}} \otimes \mathsf{cf}_{\vee}^{-1}(\mathbf{M}^{\top})^{\top})\mathbf{W}.$$

The prover extracts the trace $\mathbf{V}_{\mathsf{tr}} = \mathsf{trace}(\mathbf{V})$, re-embeds it as $\mathbf{V}_{\mathsf{emb}} = \mathsf{cf}^{-1}(\mathbf{V}_{\mathsf{tr}})$, and decomposes it into $\mathbf{Y}_{\mathsf{proj}} = \mathbf{G}_{\ell}^{-1}(\mathbf{V}_{\mathsf{emb}})$. The consistency is proven by batching and lifting the constraints to $\mathcal{R}_q$ via random tensors.

– *Sumcheck-driven Recursion.* `RoKoko` extends the sumcheck framework from SALSAA [KLOT25] to prove the well-formedness of the recursive commitments and the correctness of projections. Unlike LaBRADOR, `RoKoko` formulates and batches the structured projections as sumcheck claims, which are proven via the sumcheck protocol.

During the sumcheck step, the verifier runtime is dominated by the evaluation of the multilinear extension $\mathsf{mle}[\mathbf{M}]$ at random challenge points. To avoid dimension expansion during witness repacking, the projection dimensions ratio $\frac{m_{\mathtt{rp}}}{n_{\mathtt{rp}}}$ should align with the global shrink factor ρ, the rate at which the witness dimension is reduced, requiring $m_{\mathtt{rp}} = \rho \cdot n_{\mathtt{rp}}$. Since $\mathbf{M}$ is unstructured, evaluating $\mathsf{mle}[\mathbf{M}]$ costs the verifier $\mathcal{O}(n_{\mathtt{rp}} m_{\mathtt{rp}}) = \mathcal{O}(\rho\lambda^2)$ ring operations. This $\mathcal{O}(\rho\lambda^2)$ cost establishes a structural barrier. Setting $\rho = \mathsf{poly}(\lambda)$ reduces the recursion depth and the overall proof size, but increases the verifier time.

5.2 Modular Integration

Substitution in LaBRADOR. The LaBRADOR protocol does not natively employ the sumcheck protocol. Its shortness proof relies on unstructured Johnson-Lindenstrauss projections to bound the witness norm. Substituting the unstructured projection with the TTRP module reduces the shortness proof complexity in each recursion round to $\mathcal{O}(\lambda\mu c^2 d)$. The reduction Π^{0} transforms TTRP correctness into sumcheck claims, and the embedded sumcheck protocol in Π^{TTRP} linearises them into a polynomial evaluation claim. The terminal claim asserts that the value $w_{\mathbf{r}}$ provided by the prover equals the multilinear extension of the flattened witness at the verifier's challenge point $\mathbf{r} \in \mathcal{C}^{\log(mr)}$

$$w_{\mathbf{r}} = \mathsf{mle}[\mathsf{vec}(\mathbf{W})](\mathbf{r}) \pmod q.$$

By definition of the multilinear extension, this evaluation is an inner product. To align with the witness structure of LaBRADOR, the witness $\mathbf{W}$ is represented as r column vectors $\mathbf{w}_1, \ldots, \mathbf{w}_r \in \mathcal{R}_q^m$, assuming m and r are powers of 2. Partitioning the challenge point $\mathbf{r}$ into $\mathbf{r}_{\mathrm{col}} \in \mathcal{C}^{\log r}$ and $\mathbf{r}_{\mathrm{row}} \in \mathcal{C}^{\log m}$, and defining for a vector $\mathbf{x} = (x_1, \ldots, x_s) \in \mathcal{R}_q^s$ the structured expansion $\mathsf{tensor}(\mathbf{x}) = \big((1-x_1, x_1) \otimes \cdots \otimes (1-x_s, x_s)\big)^{\top} \in \mathcal{R}_q^{2^s}$, the expanded challenge vector decomposes via the Kronecker product

$$\mathsf{tensor}(\mathbf{r}) = \mathsf{tensor}(\mathbf{r}_{\mathrm{col}}) \otimes \mathsf{tensor}(\mathbf{r}_{\mathrm{row}}).$$

Partitioning $\mathsf{tensor}(\mathbf{r}) \in \mathcal{R}_q^{mr}$ into r contiguous blocks $\boldsymbol{\varphi}_1, \ldots, \boldsymbol{\varphi}_r \in \mathcal{R}_q^m$, where the i-th block is given by $\boldsymbol{\varphi}_i = \rho_i \cdot \mathsf{tensor}(\mathbf{r}_{\mathrm{row}})$ with $\rho_i \in \mathcal{R}_q$ as the i-th component of $\mathsf{tensor}(\mathbf{r}_{\mathrm{col}})$, the inner product decomposes as

$$\sum_{i=1}^{r} \langle \boldsymbol{\varphi}_i, \mathbf{w}_i \rangle - w_{\mathbf{r}} = 0 \pmod q.$$

LaBRADOR absorbs this claim into its constraints.

Substitution in RoKoko. The RoKoko protocol employs structured projections—specifically coarse, fine, and batch-and-lift variants—as self-reductions for the committed-linear relation Ξ^{lin} to bound witness norms. Although these native projections function as self-reductions, any such relation can alternatively be compiled into sumcheck claims. Substituting RoKoko's random projections with the Π^0 module transforms the projection directly into sumcheck claims. These claims are batched with RoKoko's internal linearization sumcheck pipeline, allowing a single execution of the sumcheck protocol per recursion round to linearise all claims.

This substitution yields three improvements. First, because RoKoko's structured projections yield large output dimensions, the prover is forced to commit to the projection image to prevent communication blowup. In contrast, TTRP projects the witness onto a small number of rows k, enabling the projected vector $\mathbf{y}_0$ to be transmitted directly in the clear. Second, avoiding projection commitments eliminates auxiliary witness components, allowing the protocol to gain larger super-constant shrink factors similar to LaBRADOR. Third, the verifier complexity of TTRP is lower by a factor $\mathcal{O}(\lambda^2/\log^2 m)$ than that of RoKoko.

6 Integration to Folding Schemes

In this section, we provide an example of integration of our TTRP range proof module Π^{TTRP} to a folding scheme. We focus our attention on the most efficient lattice-based folding scheme available Cyclo [GLLO26], and assume some familiarity.

6.1 Shortness proof of Cyclo

Cyclo is a folding scheme for R1CS relations (or its higher-degree generalization CCS [STW23]) improving on the prior work Neo [NS26]. Cyclo introduces a technique that only requires a range proof for the fresh input witnesses, but not the accumulator. Importantly, the first step of Cyclo is to run the sum-check linearization of CCS from [NS26,KS23] and obtain instance-witness pairs for a linear relation for Ajtai commitments over $\mathcal{R}_q$. In particular, this relation fits the formalism of our relation Ξ^{poly}.

The norm proof of Cyclo works in two steps. First, the fresh witnesses $\mathbf{w}_1, \ldots, \mathbf{w}_r$ with $\|\mathbf{w}_i\|_\infty < B$ for some bound B are decomposed into $\log_3(B)$ ternary subwitnesses. For each $i \in [r]$, these subwitnesses are reorganised into a single longer ternary witness, and recommitted to in this form called *extension commitment*. The consistency of the r extension commitments can be checked by the verifier. The second step is to give a sum-check-based ternary proof for the extension commitments. Combining the consistency of the commitments with the ternary proof for the extension commitments yields a tight ℓ_∞-norm proof.

The reason for creating extension commitments in Cyclo is the fact that at the time, there was no efficient succinct range RoK available.

6.2 Integration of TTRP to Cyclo

Our TTRP norm-check Π^{TTRP} acts as an ideal drop-in replacement for Cyclo's ternary range proof on extension commitments. Recall that the folded relation fits the formalism of our relation $\Xi^{\mathsf{poly}}_{f,\kappa,n,m,r,B}$, where f is the linear map that encodes the constraints, κ is the length of the commitments, n the number of constraints, m the length of the witnesses over $\mathcal{R}_q$, r the number of witnesses and finally B the Euclidean norm bound.

With the parameters that yield a 31.8 KB folding proof benchmark, the proof size Π^{TTRP} is 3.9 KB, while Cyclo's original range proof (extension commitments + sum-check) amounts to 10.4 + 3.16 KB. By integrating TTRP, the resulting proof size becomes 22.14 KB, i.e a 30% improvement.

We now estimate the ℓ_2-norm bound of the fresh witnesses to achieve the number of folding steps fixed at 64 in Cyclo's benchmark. With $\kappa \cdot \varphi = 1664$, the lattice estimator [APS15] gives us a maximum extracted Euclidean norm bound of 2^{45}. After T folding steps, the worse-case[11] norm of the accumulator grows to $\gamma T 2^{10} B$, where γ is the ℓ_2 operator norm of the challenges and 2^{10} is our slack factor. Plugging in the numbers, we obtain $B \simeq 2^{24}$.

We note that verifier efficiency–in particular the size of the challenges that are usually negligible in proof systems–is a highly relevant metric in folding schemes, due to applications such as Incrementally Verifiable Computation [Val08]. The verifier circuit of TTRP is significantly larger than that of original Cyclo. This gap is caused by the large size of the core tensors, which creates a significant overhead for applications of folding schemes in recursive proofs.

7 Parameters and Efficiency

7.1 Asymptotics

Complexity of Π^0 and Π^{TTRP}. We analyze the asymptotic efficiency of the protocols Π^0 and Π^{TTRP} under the assumption of an $\mathcal{O}(\log(mr))$-bit modulus q. Let the witness be $\mathbf{W} \in \mathcal{R}^{m \times r}$, where r denotes the split width. The ring $\mathcal{R}_q$ has degree $\varphi \in \mathcal{O}(\lambda)$. The tensor format encodes the entire witness, yielding $\mu = \lceil \log_d(mr\varphi) \rceil \in \mathcal{O}(\log(mr\varphi))$ tensor layers.

We first analyze Theorem 2 asymptotically. We set the base dimension $d \in \mathcal{O}(1)$ and the internal rank $c \in \mathcal{O}(\log \mu)$ to bound the single-row projection failure probability by a constant. The verifier runtime is dominated by evaluating the multilinear extension of the spatial tensor cores at a random point. Utilizing block-wise tensor contraction (Lemma 6), this evaluation requires

$$\mathcal{O}(k\mu_1 c^2 d) = \mathcal{O}\left(\lambda \log(mr)(\mathrm{loglog}(mr\varphi))^2\right) \in \tilde{\mathcal{O}}(\lambda)$$

ring operations. The extraction slack factor $\tilde{B}/B$ for Π^0 satisfies

$$\tilde{B}/B = \mathcal{O}\big(\sqrt{\lambda\,(\mathrm{loglog}(mr\varphi))^{\log(mr\varphi)}} \cdot (mr)^{3/2}\big) \in \mathsf{poly}(mr).$$

This satisfies the no-overflow condition $(\tilde{B}/B)^2 < q/2$ under the $\mathcal{O}(\log mr)$-bit modulus.

Next, we analyze Theorem 3. We set the threshold $\theta = \delta\eta_2$ for a constant fraction $\delta \in (0, 1)$. Substituting θ into the probability base of Eq.(15) yields

$$1 - \frac{(\eta_2 - \delta\eta_2)^2}{\eta_4 - 2\eta_2(\delta\eta_2) + (\delta\eta_2)^2} = 1 - \frac{(1-\delta)^2}{\frac{\eta_4}{\eta_2^2} - 2\delta + \delta^2}$$

To bound this base by a constant less than 1, the ratio of the moments

$$\frac{\eta_4}{\eta_2^2} = \frac{\frac{3}{4}\left(\frac{c(c+2)}{4}\right)^{\mu-1}}{\frac{1}{4}\left(\frac{c^2}{4}\right)^{\mu-1}} = 3\left(1 + \frac{2}{c}\right)^{\mu-1}$$

must be bounded by a constant. To ensure convergence, we require the internal rank to be $c \in \Omega(\mu) = \mathcal{O}(\log(mr\varphi))$. Setting $c = \mu$ yields $\lim_{\mu\to\infty} \eta_4/\eta_2^2 = 3e^2$. The verification complexity under this setting evaluates to

$$\mathcal{O}(k\mu_1 c^2 d) = \mathcal{O}(\lambda \log(mr) \log^2(mr\varphi)) \in \tilde{\mathcal{O}}(\lambda)$$

To optimize asymptotic verifier succinctness, we adopt the parameterization strategy derived from Theorem 2. The proof size and prover runtime follow this parameterization framework.

[11] Assuming that the folded witnesses are independent, the ℓ_2-norm of the accumulator would grow as $\sqrt{T}$ rather than T.

Complexity of Integration to `RoKoko`. We analyze the integration complexity using $\Pi^{\mathsf{fold\text{-}split}}$ and Π^{lin} from `RoKoko` (Table 2). Our composition is

$$\Pi^{0} \to \Pi^{\mathsf{fold\text{-}split}} \to \Pi^{\mathsf{lin}}$$

We apply TTRP for the shortness proof, proceed with the fold-and-split phase, and linearize the constraints via a batched sumcheck protocol. In `RoKoko`, the verifier complexity is bounded by $\mathcal{O}(r\lambda^2)$. Linearizing the TTRP constraints requires $\tilde{\mathcal{O}}(\lambda)$ operations for any r, which circumvents this dependence. Table 3 details the complexity comparison. The complexity of `RoKoko` is reported from the original formulation.

Protocol		**Prover**	**Verifier**	**Comm.**
$\Pi^{\mathsf{fold\text{-}split}}$	$\Xi^{\mathsf{lin}} \to \Xi^{\mathsf{sum}}$	$\mathcal{O}(mr)$	$\mathcal{O}(r/\log\lambda)$	$\mathcal{O}(1)$
Π^{lin}	$\Xi^{\mathsf{sum}} \to \Xi^{\mathsf{lin}}$	$\mathcal{O}(mr + r\lambda^2)$	$\mathcal{O}(r\lambda^2 + \log k_{\mathsf{sc}} + \log m)$	$\mathcal{O}(\log m/\log\lambda)$

Table 2. Two RoKs from [KLN+26]. We set their shrink factor $\rho := r$ to match our paper's notation.

	`RoKoko`		Ours	
$r = \lambda$	Prover	$m\lambda^2$	Prover	$m\lambda^2 \log\log(m\lambda)$
	Verifier	$\frac{\lambda^3 \log m}{\log\lambda}$	Verifier	$\frac{\lambda \log^2 m}{\log\lambda}(\log\log(m\lambda))^2$
	Comm	$\frac{\log^2 m + \lambda \log m}{\log^2 \lambda}$	Comm	$\frac{\log^2 m + \lambda \log m}{\log^2 \lambda}$
$r = m^{\epsilon}$	Prover	$m^{1+\epsilon}\lambda$	Prover	$m^{1+\epsilon}\lambda \log\log(m\lambda)$
	Verifier	$m^{\epsilon}\lambda^2$	Verifier	$\lambda \log m(\log\log(m\lambda))^2 + \frac{m^{\epsilon}}{\log\lambda}$
	Comm	$\frac{\log m + \lambda}{\log\lambda}$	Comm	$\frac{\log m + \lambda}{\log\lambda}$

Table 3. Complexity comparison of composed succinct arguments. Big-$\mathcal{O}$ notation is omitted for brevity. We use the lift-and-batch instantiation for both constructions.

7.2 Concrete Parameter Selection for TTRP

The parameter instantiation of TTRP admits multiple approaches. Following the notation in Theorem 4, we present a strategy to minimize the TT-format size[12] of the TTRP which dominate the verifier cost in Π^{TTRP} as we analyzed in Lemma 6. We fix the projection error $p \approx 2^{-90}$. For given witness lengths $m \in \{2^{20}, 2^{21}, 2^{24}\}$ over $\mathcal{R}_q$ and a predefined upper bound for the extraction slack factor $\tilde{B}/B$, we iterate over all valid TT-decompositions $m = d^{\mu}$, internal ranks c, and projection row counts k up to a threshold $k_{\max} = 300$. Within this discrete space, we discard configurations that violate the probability bounds for the modulo q overflow (Theorem 2) or the Cantelli variance concentration (Theorem 3). The search outputs the parameters that yields the minimum TT-format representation while satisfying the maximum slack constraint.

[12] We omit the polynomial degree φ here.

m	Slack	k	μ	c	Repr.
2^{20}	2^7	296	4	11	$2^{22.13}$
	2^{10}	292	4	10	$2^{21.83}$
2^{21}	2^7	296	3	7	$2^{22.41}$
	2^{10}	278	3	7	$2^{22.32}$
2^{24}	2^7	296	4	11	$2^{23.13}$
	2^{10}	292	4	10	$2^{22.83}$

Table 4. TTRP parameter instantiation. The Repr. column presents the total size of the core tensors, given by $kd((\mu-2)c^2+2c)$.

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Supplementary Material

A Omitted Proofs

A.1 Proof of theorem 4

Proof. The k' sumcheck claims evaluate the polynomials g_i over the boolean hypercube $\{0,1\}^\nu$. Each summation yields $\langle \mathbf{m}^{*(i)}, \mathsf{vec}(\overline{\mathbf{W}})\rangle - y'_i = 0 \bmod q$. For all $i \in [k']$, the k' sumcheck claims establish $\mathbf{y}' = \mathbf{\Gamma} \cdot \mathbf{M} \cdot \mathsf{vec}(\overline{\mathbf{W}}) \bmod q$.
Correctness. Consider an instance $(\mathbf{T}, \mathbf{W}) \in \Xi^{\mathsf{poly}}$. The j-th row of $\mathbf{y}$ is evaluated as

$$\mathbf{y}^{(j)} = \mathbf{M}^{(j)} \cdot \mathsf{vec}(\overline{\mathbf{W}}) \bmod q = \sum_{h=1}^{mr} \mathbf{M}_h^{(j)} \overline{\mathbf{W}_h} \bmod q.$$

Taking the constant term of $\mathbf{y}^{(j)}$ and utilizing the property of power-of-two cyclotomic rings where $\mathsf{ct}(a\bar{b}) = \langle \mathsf{cf}(a), \mathsf{cf}(b)\rangle$, we obtain

$$\mathbf{y}_0^{(j)} = \sum_{h=1}^{mr} \mathsf{ct}\left(\mathbf{M}_h^{(j)} \overline{\mathbf{W}_h}\right) = \sum_{h=1}^{mr} \langle \mathsf{cf}\left(\mathbf{M}_h^{(j)}\right), \mathsf{cf}(\mathbf{W}_h)\rangle = \mathbf{M}_{\mathbb{Z}}^{(j)} \cdot \mathsf{cf}(\mathsf{vec}(\mathbf{W})) \pmod q.$$

Given $\|\mathbf{W}\|_2 < B$, the expected squared norm of $\mathbf{y}_0$ is bounded by

$$\mathbb{E}\left[\|\mathbf{y}_0\|_2^2\right] = kc^{\mu-1}\left(\frac{1}{2}\right)^{\mu} \|\mathsf{cf}(\mathsf{vec}(\mathbf{W}))\|_2^2 < kc^{\mu-1}\left(\frac{1}{2}\right)^{\mu} B^2.$$

Since $\|\mathbf{y}_0\|_2^2$ is a non-negative random variable, applying Markov's inequality yields $\Pr\left[\|\mathbf{y}_0\|_2^2 \geqslant 2\mathbb{E}\left[\|\mathbf{y}_0\|_2^2\right]\right] \leqslant \frac{1}{2}$. An honest prover passes the check $\|\mathbf{y}_0\|_2 \leqslant \sqrt{k\left(\frac{c}{2}\right)^{\mu-1}} \cdot B$ with probability at least $1/2$. The identity check holds as

$$\mathsf{ct}(\mathbf{y}') = \mathsf{ct}(\mathbf{\Gamma} \cdot \mathbf{y}) = \mathbf{\Gamma} \cdot \mathsf{ct}(\mathbf{y}) = \mathbf{\Gamma} \cdot \mathbf{y}_0.$$

Knowledge Soundness. Let $\hat{B} := \sqrt{k\,(c/2)^{\mu-1}}B$. Since the polynomial map constraint $f(\mathbf{W}) = 0$ is preserved in the reduction, we omit it from the following proof for simplicity. We split the protocol into two phases and analyze them separately.

a) The tensor train random projection phase $\mathit{\Pi}_{\mathtt{TT}}^{\mathsf{0}}$ which reduces an instance $\big(\mathbf{T}, (\mathbf{W}, \mathbf{S})\big)$ to an instance with additional norm and constant term checks on the projected witness

$$\big[\mathbf{T}, (\mathbf{W}, \mathbf{S})\big] \mapsto \big[\big(\mathbf{T}, (\mathbf{W}, \mathbf{S})\big) \text{ and } \mathsf{ct}(\mathbf{M} \cdot \mathsf{vec}(\overline{\mathbf{W}})) = \mathbf{y}_0 \text{ and } \|\mathbf{y}_0\|_2 \leqslant \hat{B}\big].$$

This corresponds to the first challenge-response, denoted by $\langle \mathcal{P}_{\mathtt{TT}}^*, \mathcal{V}_{\mathtt{TT}}\rangle$.

b) The aggregation phase Π^0_{SZ} which reduces an instance as

$$\big[\big(\mathbf{T},(\mathbf{W},\mathbf{S})\big) \quad \text{and} \quad \mathsf{ct}(\mathbf{M}\cdot\mathsf{vec}(\overline{\mathbf{W}}))=\mathbf{y}_0\big] \mapsto \big[\big(\mathbf{T},(\mathbf{W},\mathbf{S})\big) \quad \text{and} \quad \mathbf{\Gamma}\cdot\mathbf{M}\cdot\mathsf{vec}(\overline{\mathbf{W}})=\mathbf{y}'\big].$$

This corresponds to the ν-th challenge-response, denoted by $\langle \mathcal{P}^*_{\mathsf{SZ}}, \mathcal{V}_{\mathsf{SZ}}\rangle$.

Extractor $\mathcal{E}_{\mathsf{SZ}}$. W.L.O.G., assume that the prover $\mathcal{P}^*_{\mathsf{SZ}}$ is deterministic. $\mathcal{E}_{\mathsf{SZ}}$ runs the prover $\mathcal{P}^*_{\mathsf{SZ}}$ on some challenges $\mathbf{\Gamma} \leftarrow \mathbb{Z}_q^{k'\times k}$ until it obtains $(\mathbf{y}', \mathbf{W}, \mathbf{S})$ satisfying the constraints. If the prover fails, then abort.

$$\begin{aligned} \mathbf{A}\cdot\mathbf{W} &= \mathbf{T} \bmod q, & \|\mathbf{S}\|_{\mathsf{op}} &\leqslant \varrho,\\ \mathbf{\Gamma}\cdot\mathbf{M}\cdot\mathsf{vec}(\overline{\mathbf{W}}) &= \mathbf{y}' \bmod q, & \|\mathbf{WS} \bmod q\|_2 &\leqslant B',\\ \mathsf{ct}(\mathbf{y}') &= \mathbf{\Gamma}\cdot\mathbf{y}_0 \bmod q. \end{aligned}$$

If $\mathsf{ct}(\mathbf{M}\cdot\mathsf{vec}(\overline{\mathbf{W}})) = \mathbf{y}_0 \bmod q$, $\mathcal{E}_{\mathsf{SZ}}$ outputs $(\mathbf{W},\mathbf{S})$ and concludes the extraction. Otherwise, $\mathcal{E}_{\mathsf{SZ}}$ re-runs the prover $\mathcal{P}^*_{\mathsf{SZ}}$ on new challenges $\mathbf{\Gamma}'$ to obtain $(\mathbf{y}'', \mathbf{W}', \mathbf{S}')$ satisfying

$$\begin{aligned} \mathbf{A}\cdot\mathbf{W}' &= \mathbf{T} \bmod q, & \|\mathbf{S}'\|_\infty &\leqslant \varrho,\\ \mathbf{\Gamma}'\cdot\mathbf{M}\cdot\mathsf{vec}(\overline{\mathbf{W}'}) &= \mathbf{y}'' \bmod q, & \|\mathbf{W}'\mathbf{S}' \bmod q\|_2 &\leqslant B',\\ \mathsf{ct}(\mathbf{y}'') &= \mathbf{\Gamma}'\cdot\mathbf{y}_0 \bmod q. \end{aligned}$$

If $\mathbf{W} \neq \mathbf{W}' \bmod q$, then $\mathbf{WSS}' - \mathbf{W}'\mathbf{S}'\mathbf{S} \neq 0 \bmod q$. This yields $\mathbf{A}(\mathbf{WSS}' - \mathbf{W}'\mathbf{S}'\mathbf{S}) = 0$ and $\|\mathbf{WSS}' - \mathbf{W}'\mathbf{S}'\mathbf{S} \bmod q\|_2 \leqslant 2B'\varrho$. Any non-zero column $\mathbf{x}'$ of this difference satisfies $(\mathbf{A},\mathbf{x}') \in \Xi^{\mathsf{sis}}_{2B'\varrho}$.

If $\mathbf{W} = \mathbf{W}' \bmod q$ and the trace conditions diverge

$$\begin{aligned} \mathsf{ct}(\mathbf{M}\cdot\mathsf{vec}(\overline{\mathbf{W}})) &\neq \mathbf{y}_0 \bmod q,\\ \mathsf{ct}(\mathbf{\Gamma}'\cdot\mathbf{M}\cdot\mathsf{vec}(\overline{\mathbf{W}})) &= \mathbf{\Gamma}'\cdot\mathbf{y}_0 \bmod q, \end{aligned}$$

since $\mathbf{\Gamma}'$ is over $\mathbb{Z}_q$, we have $\mathbf{\Gamma}'\cdot\mathsf{ct}(\mathbf{M}\cdot\mathsf{vec}(\overline{\mathbf{W}})) = \mathbf{\Gamma}'\cdot\mathbf{y}_0 \bmod q$. Thus, the vector $\mathbf{\Gamma}'\cdot\big(\mathsf{ct}(\mathbf{M}\cdot\mathsf{vec}(\overline{\mathbf{W}})) - \mathbf{y}_0\big) \bmod q$ is an evaluation of k' independent random linear combinations. By the Schwartz-Zippel lemma, if the constant vector $\mathsf{ct}(\mathbf{M}\cdot\mathsf{vec}(\overline{\mathbf{W}})) - \mathbf{y}_0$ is non-zero, this evaluation yields $\mathbf{0}$ with probability at most $q^{-k'}$.

Extractor $\mathcal{E}_{\mathsf{TT}}$. W.L.O.G., assume that the prover $\mathcal{P}^*_{\mathsf{TT}}$ is deterministic. $\mathcal{E}_{\mathsf{TT}}$ runs the prover $\mathcal{P}^*_{\mathsf{TT}}$ on some challenges $\{\mathcal{M}_i^{(j)}\}_{i\in[\mu],j\in[k]}$ until it obtains $(\mathbf{y}_0, \mathbf{W}, \mathbf{S})$ satisfying the constraints. If the prover fails, then abort.

$$\begin{aligned} \mathbf{A}\cdot\mathbf{W} &= \mathbf{T} \bmod q, & \|\mathbf{S}\|_{\mathsf{op}} &\leqslant \varrho,\\ \mathsf{ct}(\mathbf{M}\cdot\mathsf{vec}(\overline{\mathbf{W}})) &= \mathbf{y}_0 \bmod q, & \|\mathbf{W}\cdot\mathbf{S} \bmod q\|_2 &\leqslant B',\\ & & \|\mathbf{y}_0\|_2 &\leqslant \hat{B}. \end{aligned}$$

If $\|\mathbf{W}\|_2 \leqslant \tilde{B}$, then $\mathcal{E}_{\mathsf{TT}}$ outputs $(\mathbf{W},\mathbf{S})$ and concludes the extraction. If $\|\mathbf{W}\|_2 > \tilde{B}$, then $\mathcal{E}_{\mathsf{TT}}$ re-runs the prover $\mathcal{P}^*_{\mathsf{TT}}$ on new challenges $\{\mathcal{M}_i'^{(j)}\}$ to obtain $(\mathbf{y}_0', \mathbf{W}', \mathbf{S}')$ bounded similarly.

As before, if $\mathbf{W} \neq \mathbf{W}' \bmod q$, we extract a SIS solution $\mathbf{x}' \in \Xi^{\mathsf{sis}}_{2B'\varrho}$. Assuming $\mathbf{W} = \mathbf{W}' \bmod q$, the verifier's ℓ_2 norm check ensures that $\|\mathbf{y}_0\|_2 \leqslant \hat{B}$. Since $\mathbf{M}_{\mathbb{Z}}$ (and its embedding $\mathbf{M}$) is independent of $\mathbf{W}$, by Theorem 3 and the established equivalence $\mathbf{y}_0 = \mathbf{M}_{\mathbb{Z}}\cdot\mathsf{cf}(\mathsf{vec}(\mathbf{W}))$, we have

$$\|\mathbf{W}\|_2 \leqslant \frac{\hat{B}}{\sqrt{\theta}} = \frac{\sqrt{k(c/2)^{\mu-1}}}{\sqrt{\theta}}\cdot B = \tilde{B}$$

unless with a probability of at most p. A successful extraction outputs $(\mathbf{W},\mathbf{S})$ satisfying the starting relation or breaks SIS. The total knowledge error is at most $\delta_0 = p + q^{-k'}$.

A.2 Proof of Lemma 3

Proof. We only prove the Eq.(11) here. The proof for the Eq.(12) is given in Appendix A.3. Let $\mathbf{p}^\top \in \mathbb{Z}^{1\times m}$ be the random projection row defined by $\mathbf{p}^\top := \mathsf{Mat}\left(\mathcal{M}_1 \otimes \cdots \otimes \mathcal{M}_\mu\right)$. The expected squared norm of the projection is $\mathbb{E}[(\mathbf{p}^\top\mathbf{x})^2] = \mathbb{E}[\mathbf{x}^\top\mathbf{p}\mathbf{p}^\top\mathbf{x}] = \mathbf{x}^\top\mathbb{E}[\mathbf{p}\mathbf{p}^\top]\mathbf{x}$. Applying Fact 1 sequentially, we decompose $\mathbf{p}^\top$ as

$$\mathbf{p}^\top = \mathbf{M}_1\left(\mathbf{I}_d \otimes \mathbf{M}_2\right)\cdots\left(\mathbf{I}_{d^{\mu-1}} \otimes \mathbf{M}_\mu\right), \tag{19}$$

where $\mathbf{M}_i := \mathsf{Mat}(\mathcal{M}_i)$. Let $\sigma^2 := \mathsf{Var}(\mathcal{D}_{\mathsf{ghl}}) = 1/2$ be the variance of $\mathcal{D}_{\mathsf{ghl}}$. Since each $\mathbf{M}_i$ consists of independent entries with zero mean and variance σ^2, we evaluate the expectations of their Gram matrices

$$\begin{aligned}
\mathbb{E}[\mathbf{M}_1^\top\mathbf{M}_1] &= \sigma^2\mathbf{I}_{cd},\\
\mathbb{E}[\mathbf{M}_i^\top\mathbf{M}_i] &= c\sigma^2\mathbf{I}_{cd} \quad \text{for } 1 < i < \mu,\\
\mathbb{E}[\mathbf{M}_\mu^\top\mathbf{M}_\mu] &= c\sigma^2\mathbf{I}_d.
\end{aligned}$$

We define the partial product row vector $\mathbf{p}_i^\top := \mathbf{M}_1\left(\mathbf{I}_d \otimes \mathbf{M}_2\right)\cdots\left(\mathbf{I}_{d^{i-1}} \otimes \mathbf{M}_i\right) \in \mathbb{Z}^{1\times cd^i}$ and prove by induction that $\mathbb{E}[\mathbf{p}_i\mathbf{p}_i^\top] = c^{i-1}\sigma^{2i}\mathbf{I}_{cd^i}$ for $1 \leqslant i < \mu$. The base case ($i = 1$) holds trivially. Assume the hypothesis holds for $i-1$. Since $\mathbf{M}_i$ is sampled independently of the partial product $\mathbf{p}_{i-1}$, we have

$$\begin{aligned}
\mathbb{E}\left[\mathbf{p}_i\mathbf{p}_i^\top\right] &= \mathbb{E}_{\mathbf{M}_i}\left[\left(\mathbf{I}_{d^{i-1}} \otimes \mathbf{M}_i^\top\right)\mathbb{E}_{\mathbf{p}_{i-1}}\left[\mathbf{p}_{i-1}\mathbf{p}_{i-1}^\top\right]\left(\mathbf{I}_{d^{i-1}} \otimes \mathbf{M}_i\right)\right]\\
&= \mathbb{E}_{\mathbf{M}_i}\left[\left(\mathbf{I}_{d^{i-1}} \otimes \mathbf{M}_i^\top\right)\left(c^{i-2}\sigma^{2(i-1)}\mathbf{I}_{cd^{i-1}}\right)\left(\mathbf{I}_{d^{i-1}} \otimes \mathbf{M}_i\right)\right]\\
&= c^{i-2}\sigma^{2(i-1)}\,\mathbb{E}_{\mathbf{M}_i}\left[\left(\mathbf{I}_{d^{i-1}} \otimes \mathbf{M}_i^\top\right)\left(\mathbf{I}_{d^{i-1}} \otimes \mathbf{I}_c\right)\left(\mathbf{I}_{d^{i-1}} \otimes \mathbf{M}_i\right)\right]\\
&= c^{i-2}\sigma^{2(i-1)}\,\mathbb{E}_{\mathbf{M}_i}\left[\mathbf{I}_{d^{i-1}} \otimes \left(\mathbf{M}_i^\top\mathbf{M}_i\right)\right]\\
&= c^{i-2}\sigma^{2(i-1)}\left(\mathbf{I}_{d^{i-1}} \otimes c\sigma^2\mathbf{I}_{cd}\right) = c^{i-1}\sigma^{2i}\mathbf{I}_{cd^i}.
\end{aligned}$$

For the final term ($i = \mu$), applying the same decoupling strategy to $\mathbf{p}^\top = \mathbf{p}_{\mu-1}^\top\left(\mathbf{I}_{d^{\mu-1}} \otimes \mathbf{M}_\mu\right)$ yields

$$\begin{aligned}
\mathbb{E}\left[\mathbf{p}\mathbf{p}^\top\right] &= \mathbb{E}_{\mathbf{M}_\mu}\left[\left(\mathbf{I}_{d^{\mu-1}} \otimes \mathbf{M}_\mu^\top\right)\mathbb{E}_{\mathbf{p}_{\mu-1}}\left[\mathbf{p}_{\mu-1}\mathbf{p}_{\mu-1}^\top\right]\left(\mathbf{I}_{d^{\mu-1}} \otimes \mathbf{M}_\mu\right)\right]\\
&= c^{\mu-2}\sigma^{2(\mu-1)}\left(\mathbf{I}_{d^{\mu-1}} \otimes \mathbb{E}[\mathbf{M}_\mu^\top\mathbf{M}_\mu]\right)\\
&= c^{\mu-2}\sigma^{2(\mu-1)}\left(\mathbf{I}_{d^{\mu-1}} \otimes c\sigma^2\mathbf{I}_d\right) = c^{\mu-1}\sigma^{2\mu}\mathbf{I}_{d^\mu}.
\end{aligned}$$

that $d^\mu = m$ and $\sigma^2 = 1/2$, the expected matrix simplifies to $\mathbb{E}[\mathbf{p}\mathbf{p}^\top] = \frac{c^{\mu-1}}{2^\mu}\mathbf{I}_m$. Substituting this back into our initial formulation yields $\mathbb{E}[(\mathbf{p}^\top\mathbf{x})^2] = \frac{c^{\mu-1}}{2^\mu}\|\mathbf{x}\|_2^2$.

A.3 The Variance

We can establish the variance bound of the TTRP over the $\mathcal{D}_{\mathsf{ghl}}$ distribution by adapting the indicator contraction method from [RR21]. By replacing Rademacher moments with $\mathcal{D}_{\mathsf{ghl}}$ moments. From now on, we use m_i to denoted the i-th moment of $\mathcal{D}_{\mathsf{ghl}}$, i.e., $m_1 = m_3 = 0$ and $m_2 = m_4 = 1/2$

Lemma 7 (4-th Moment of Vector Inner Product). *Let $\mathbf{a} \in \mathbb{Z}^m$ be a random vector with i.i.d. entries drawn from $\mathcal{D}_{\mathsf{ghl}}$, and let $\mathbf{b} \in \mathbb{Z}^m$ be a fixed vector. Then*

$$\mathbb{E}[\langle\mathbf{a},\mathbf{b}\rangle^4] \leqslant \frac{3}{4}\|\mathbf{b}\|_2^4$$

Proof. Since the odd moments of $\mathcal{D}_{\mathsf{ghl}}$ are 0, following the expansion of non-zero summands, we have

$$\mathbb{E}[\langle \mathbf{a}, \mathbf{b}\rangle^4] = \sum_{i\in[m]} \mathbb{E}[\mathbf{a}_i^4]\mathbf{b}_i^4 + 3\sum_{i\neq j} \mathbb{E}[\mathbf{a}_i^2]\mathbb{E}[\mathbf{a}_j^2]\mathbf{b}_i^2\mathbf{b}_j^2$$

Substituting the moments m_2 and m_4, we get

$$\mathbb{E}[\langle \mathbf{a}, \mathbf{b}\rangle^4] = \frac{1}{2}\sum_i \mathbf{b}_i^4 + \frac{3}{4}\sum_{i\neq j}\mathbf{b}_i^2\mathbf{b}_j^2 = \frac{3}{4}\left(\sum_i \mathbf{b}_i^2\right)^2 - \frac{1}{4}\sum_i \mathbf{b}_i^4$$

Since $\sum_i \mathbf{b}_i^4 \geqslant 0$, we get the upper bound $\mathbb{E}[\langle \mathbf{a}, \mathbf{b}\rangle^4] \leqslant \frac{3}{4}\|\mathbf{b}\|_2^4$.

Lemma 8 (4-th Moment of Matrix Product). *Let $\mathbf{A} \in \mathbb{Z}^{d\times c}$ be a random matrix with i.i.d. entries drawn from $\mathcal{D}_{\mathsf{ghl}}$, and let $\mathbf{B} \in \mathbb{Z}^{p\times d}$ be a fixed matrix. Then*

$$\mathbb{E}[\|\mathbf{BA}\|_F^4] \leqslant \frac{c(c+2)}{4}\|\mathbf{B}\|_F^4$$

Proof. Let $\mathbf{M} = \mathbf{B}^\top\mathbf{B}$. The expansion of $\mathbb{E}[\|\mathbf{BA}\|_F^4] = \mathbb{E}[\langle \mathbf{M}, \mathbf{AA}^\top\rangle^2]$ begins by converting the Frobenius norm into the trace form. By definition, $\|\mathbf{BA}\|_F^2 = \mathrm{tr}(\mathbf{A}^\top\mathbf{MA}) = \sum_{j\in[c]}\sum_{i_1,i_2\in[d]} \mathbf{A}_{i_1 j}\mathbf{M}_{i_1 i_2}\mathbf{A}_{i_2 j}$. Squaring this term and applying the linearity of expectation yields a joint summation over six indices

$$\mathbb{E}[\|\mathbf{BA}\|_F^4] = \sum_{i_1,i_2,i_3,i_4\in[d]}\ \sum_{j,k\in[c]} \mathbb{E}[\mathbf{A}_{i_1 j}\mathbf{A}_{i_2 j}\mathbf{A}_{i_3 k}\mathbf{A}_{i_4 k}]\mathbf{M}_{i_1 i_2}\mathbf{M}_{i_3 i_4}$$

Because the entries of $\mathbf{A}$ are independent and have mean zero, the expectation $\mathbb{E}[\mathbf{A}_{i_1 j}\mathbf{A}_{i_2 j}\mathbf{A}_{i_3 k}\mathbf{A}_{i_4 k}]$ evaluates to zero if any row index appears exactly once. To yield a non-zero expectation, the row indices must either be fully identical or paired. This partitions the non-zero outcomes into four mutually exclusive cases

$$\begin{aligned}
\mathbb{E}[\|\mathbf{BA}\|_F^4] = &\sum_{i\in[d]}\sum_{j,k\in[c]} \mathbb{E}[\mathbf{A}_{ij}^2\mathbf{A}_{ik}^2]\mathbf{M}_{ii}^2 \\
&+ \sum_{\substack{i_1,i_3\in[d]\\ i_1\neq i_3}}\sum_{j,k\in[c]} \mathbb{E}[\mathbf{A}_{i_1 j}^2\mathbf{A}_{i_3 k}^2]\mathbf{M}_{i_1 i_1}\mathbf{M}_{i_3 i_3} \\
&+ \sum_{\substack{i_1,i_2\in[d]\\ i_1\neq i_2}}\sum_{j,k\in[c]} \mathbb{E}[\mathbf{A}_{i_1 j}\mathbf{A}_{i_2 j}\mathbf{A}_{i_1 k}\mathbf{A}_{i_2 k}]\mathbf{M}_{i_1 i_2}^2 \\
&+ \sum_{\substack{i_1,i_2\in[d]\\ i_1\neq i_2}}\sum_{j,k\in[c]} \mathbb{E}[\mathbf{A}_{i_1 j}\mathbf{A}_{i_2 j}\mathbf{A}_{i_2 k}\mathbf{A}_{i_1 k}]\mathbf{M}_{i_1 i_2}\mathbf{M}_{i_2 i_1}
\end{aligned}$$

Evaluating these summations under m_2 and m_4 yields

1. For $i_1 = i_2 = i_3 = i_4$: $\left(c\cdot\frac{1}{2} + c(c-1)\cdot\frac{1}{4}\right)\sum_i \mathbf{M}_{ii}^2 = \frac{c(c+1)}{4}\sum_i \mathbf{M}_{ii}^2$.
2. For $i_1 = i_2 \neq i_3 = i_4$: $c^2\cdot\frac{1}{4}\sum_{i\neq\ell}\mathbf{M}_{ii}\mathbf{M}_{\ell\ell} = \frac{c^2}{4}\left(\mathrm{tr}(\mathbf{M})^2 - \sum_i \mathbf{M}_{ii}^2\right)$.
3. For $i_1 = i_3 \neq i_2 = i_4$: $c\cdot\frac{1}{4}\sum_{i\neq\ell}\mathbf{M}_{i\ell}^2 = \frac{c}{4}\left(\|\mathbf{M}\|_F^2 - \sum_i \mathbf{M}_{ii}^2\right)$.
4. For $i_1 = i_4 \neq i_2 = i_3$: $c\cdot\frac{1}{4}\sum_{i\neq\ell}\mathbf{M}_{i\ell}\mathbf{M}_{\ell i} = \frac{c}{4}\left(\|\mathbf{M}\|_F^2 - \sum_i \mathbf{M}_{ii}^2\right)$.

Summing all terms together

$$\begin{aligned}
\mathbb{E}[\|\mathbf{BA}\|_F^4] &= \frac{c^2}{4}\mathrm{tr}(\mathbf{M})^2 + \frac{c}{2}\|\mathbf{M}\|_F^2 + \left(\frac{c^2+c}{4} - \frac{c^2}{4} - \frac{c}{2}\right)\sum_i \mathbf{M}_{ii}^2 \\
&= \frac{c^2}{4}\|\mathbf{B}\|_F^4 + \frac{c}{2}\|\mathbf{B}^\top\mathbf{B}\|_F^2 - \frac{c}{4}\sum_i \mathbf{M}_{ii}^2
\end{aligned}$$

Using $\|\mathbf{B}^\top\mathbf{B}\|_F^2 \leqslant \|\mathbf{B}\|_F^4$ and dropping the negative term, we obtain $\mathbb{E}[\|\mathbf{BA}\|_F^4] \leqslant \frac{c(c+2)}{4}\|\mathbf{B}\|_F^4$.

Lemma 9 (Variance). *Let $y = f_{\mathsf{TT}(\mu,c)}(\mathbf{x})$ be the single-row output of the TTRP as defined in Definition 4. For any input $\mathbf{x} \in \mathbb{Z}^{m\varphi}$, we have*

$$\mathbb{E}[y^4] \leqslant \frac{3}{4}\left(\frac{c(c+2)}{4}\right)^{\mu-1}\|\mathbf{x}\|_2^4,$$

$$\mathsf{Var}(y^2) \leqslant \left(\mathbb{E}[y^2]\right)^2\left[3\left(1+\frac{2}{c}\right)^{\mu-1}-1\right]$$

where $\mathbb{E}[y^2] = \frac{c^{\mu-1}}{2^\mu}\|\mathbf{x}\|_2^2$.

Proof. The projection $y = \mathsf{Mat}(\mathcal{M}_1 \otimes \cdots \otimes \mathcal{M}_\mu)\cdot\mathbf{x}$ can be unrolled from right to left as a sequence of matrix-matrix and vector-matrix products. Let $\mathbf{Y}_\mu = \mathbf{AB}$ where $\mathbf{A} = \mathsf{Mat}(\mathcal{M}_\mu) \in \mathbb{Z}^{c\times d\varphi}$ and $\mathbf{B} = [\mathbf{x}_1, \ldots, \mathbf{x}_{d^{\mu-1}}] \in \mathbb{Z}^{d\varphi\times d^{\mu-1}}$. By applying Lemma 8 iteratively for $\mu, \mu-1, \ldots, 2$ (a total of $\mu-1$ times), the 4-th moment scales by $\frac{c(c+2)}{4}$ at each layer. Finally, $\mathcal{M}_1 \in \mathbb{Z}^{1\times dc}$ acts as a row vector, so applying Lemma 7 yields the final bounding factor of $\frac{3}{4}$

$$\mathbb{E}[y^4] \leqslant \frac{3}{4}\left(\frac{c(c+2)}{4}\right)^{\mu-1}\|\mathbf{x}\|_2^4$$

The variance is then bounded by

$$\begin{aligned}\mathrm{Var}(y^2) = \mathbb{E}[y^4] - (\mathbb{E}[y^2])^2 &\leqslant \frac{3}{4}\left(\frac{c(c+2)}{4}\right)^{\mu-1}\|\mathbf{x}\|_2^4 - \left(\mathbb{E}[y^2]\right)^2 \\ &= (\mathbb{E}[y^2])^2\left[3\left(1+\frac{2}{c}\right)^{\mu-1}-1\right].\end{aligned}$$